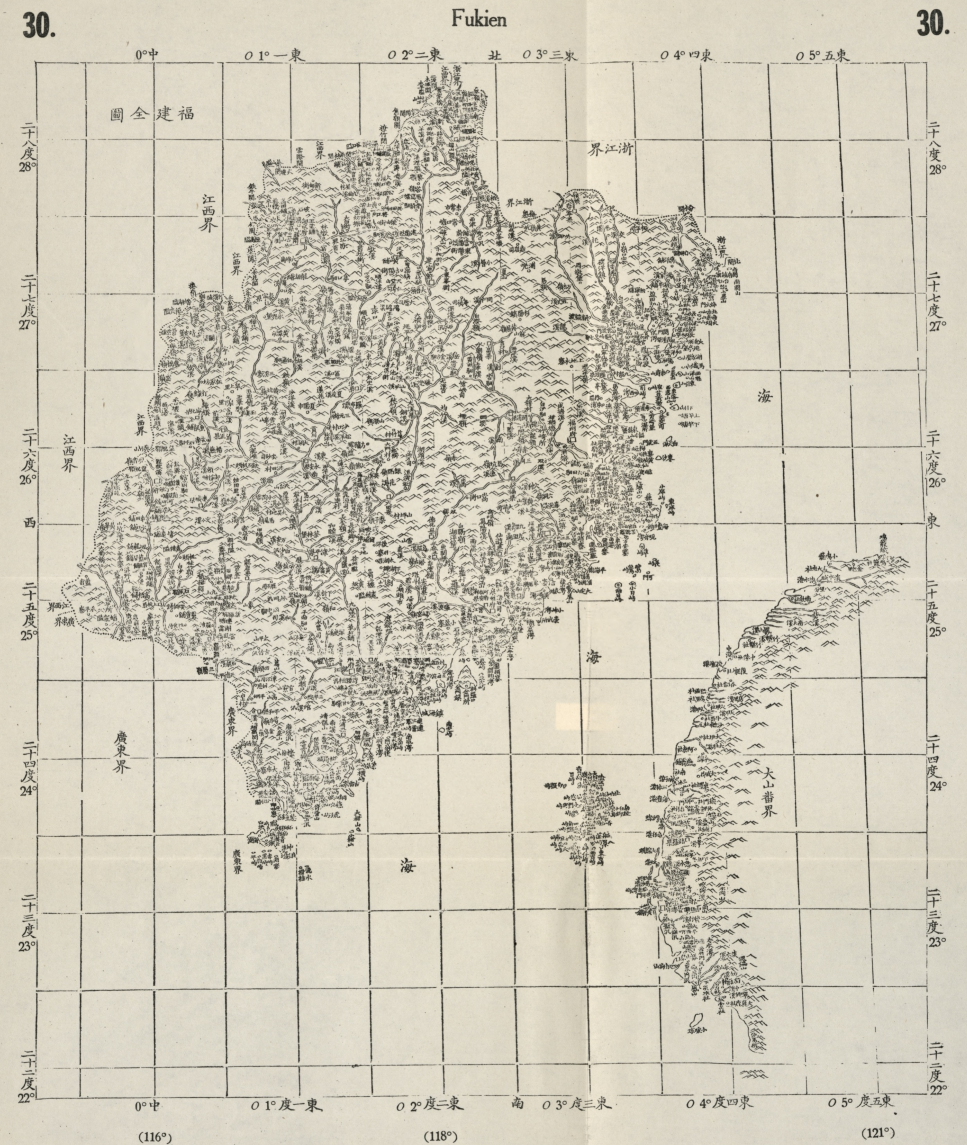
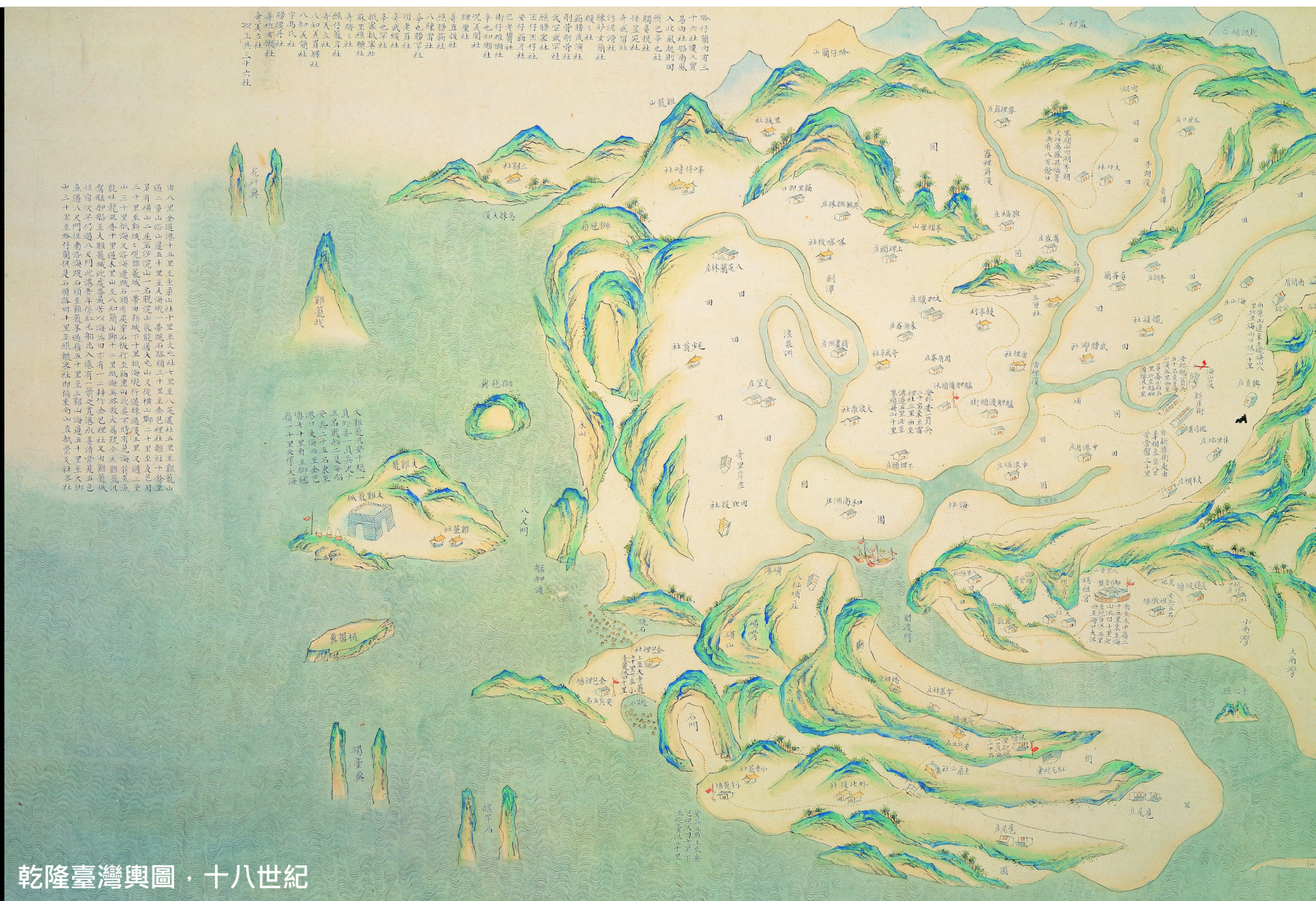


Map Spotlight 3: Map Projection

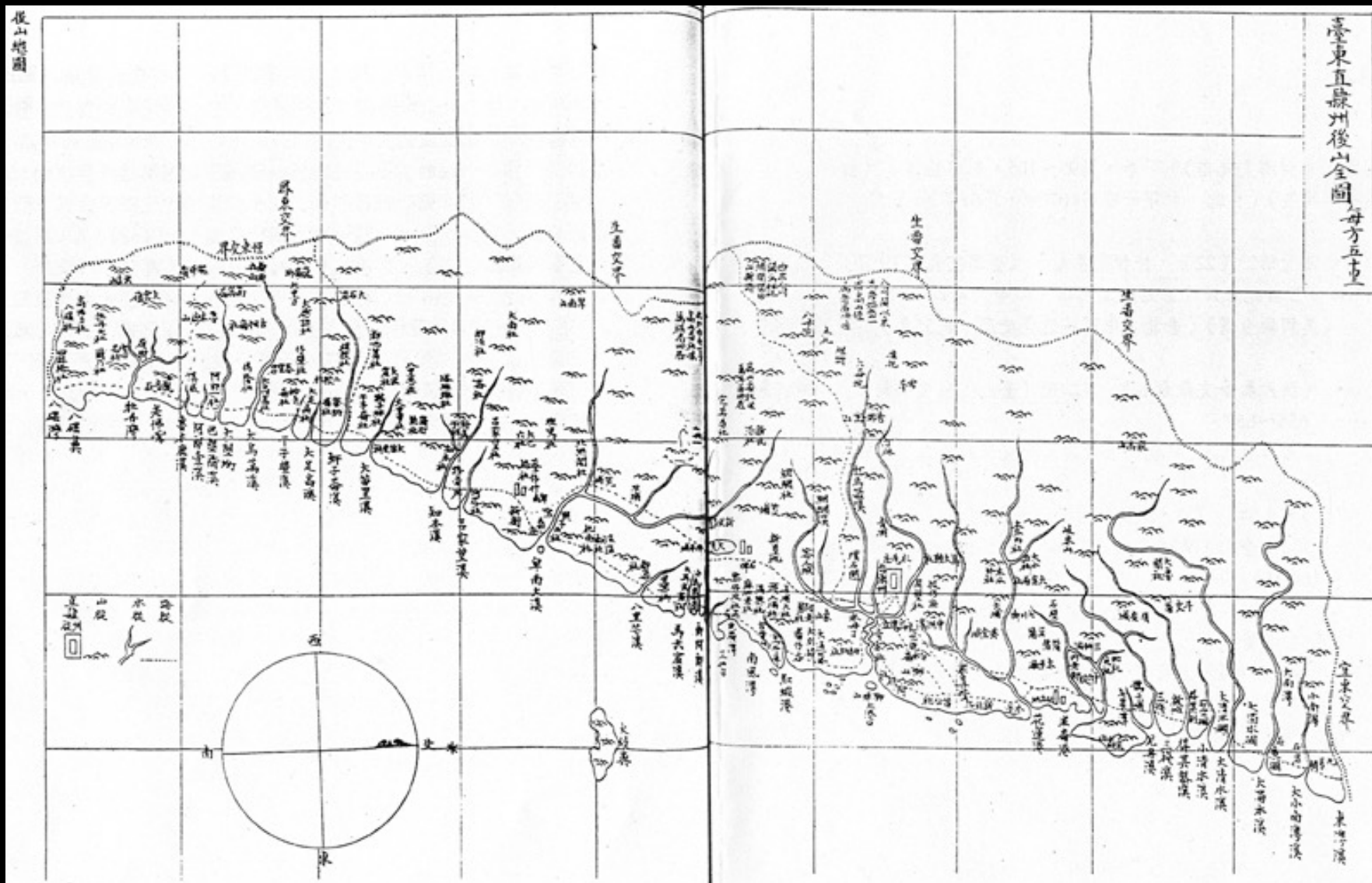
清康熙皇輿全覽圖 · 1708-1718



山水畫法



計里畫方法



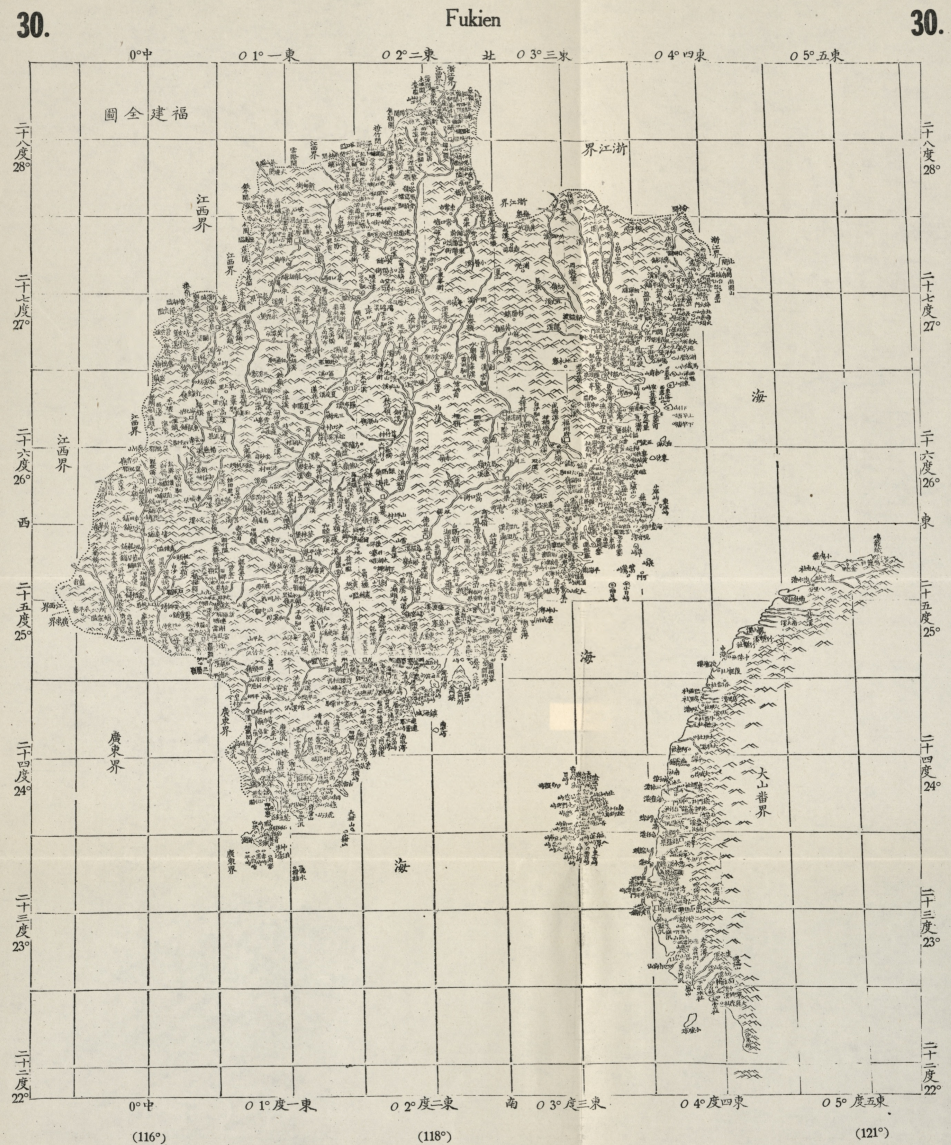
台東直隸州後山全圖，1887

丈量山水併用法

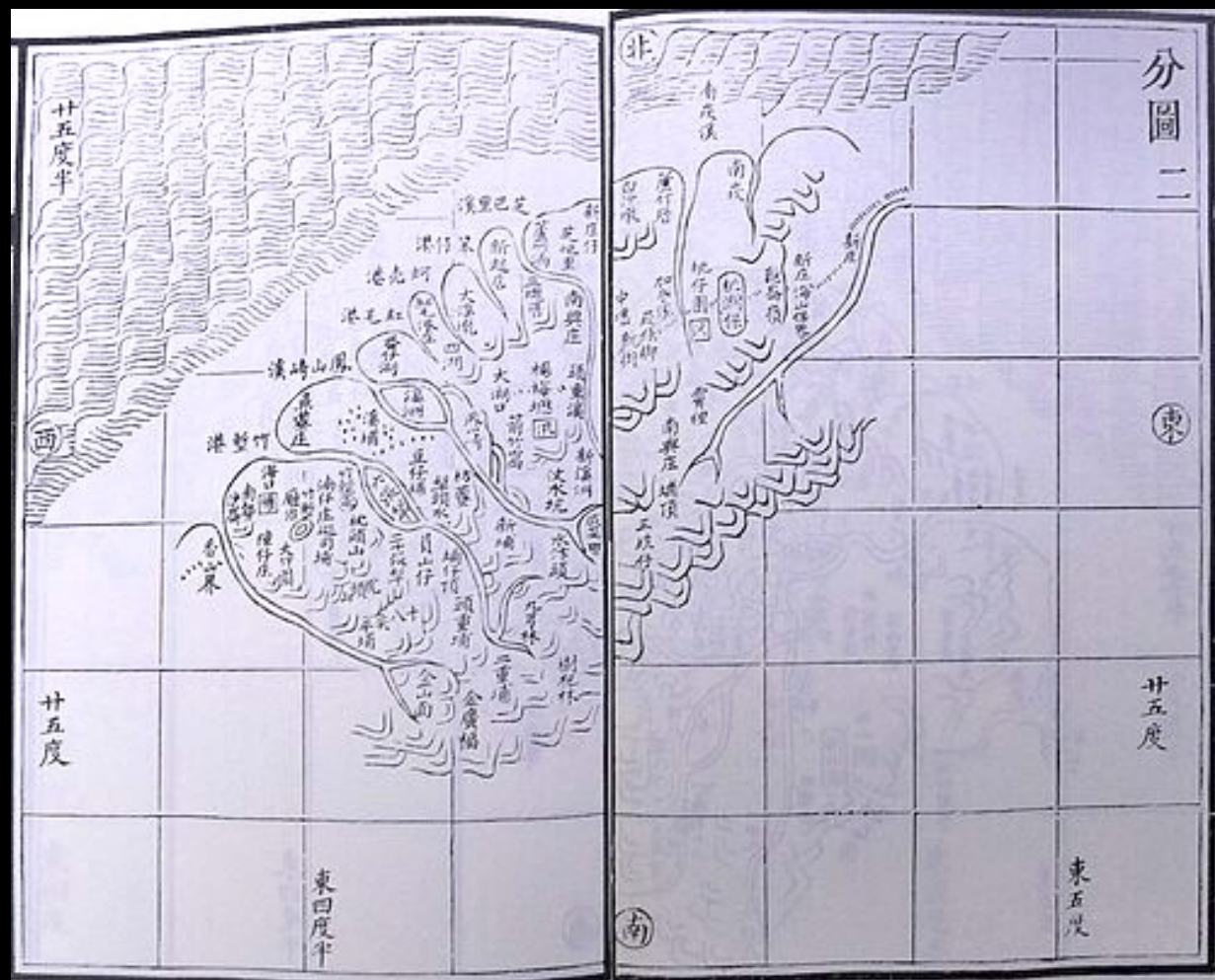


經緯度三角測量法

清康熙皇輿全覽圖，1708-1718



經緯度計里畫方併用法

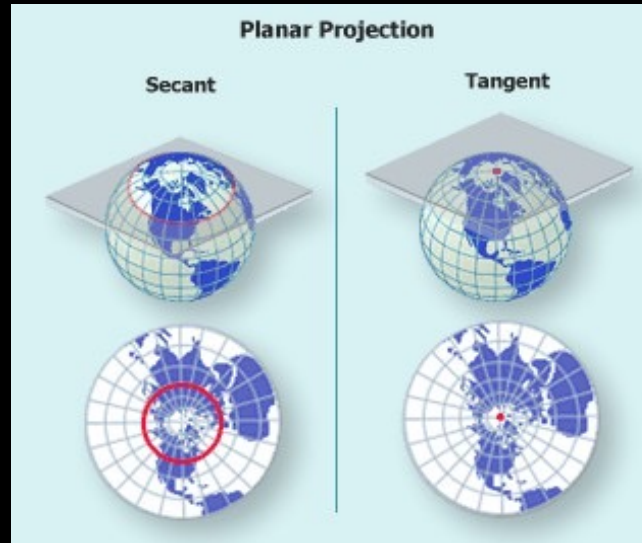


淡水廳沿海礁沙形勢圖，1871

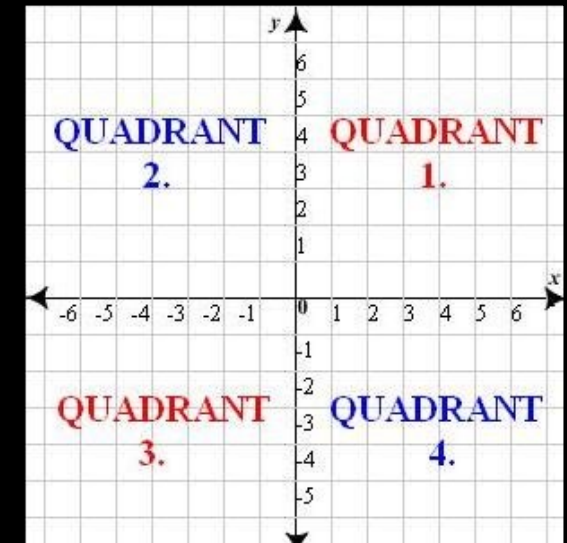
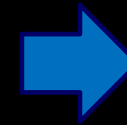
Key Concept



Geographic
Coordinate System
Latitude and Longitude



Map
Projection



Plane
Coordinate
X-Y Coordinate

完整的投影坐標系統命名：Datum + Projection，例如：TWD 97- TM2

Why Does Map Projection matter?

Mercator Projection World Map

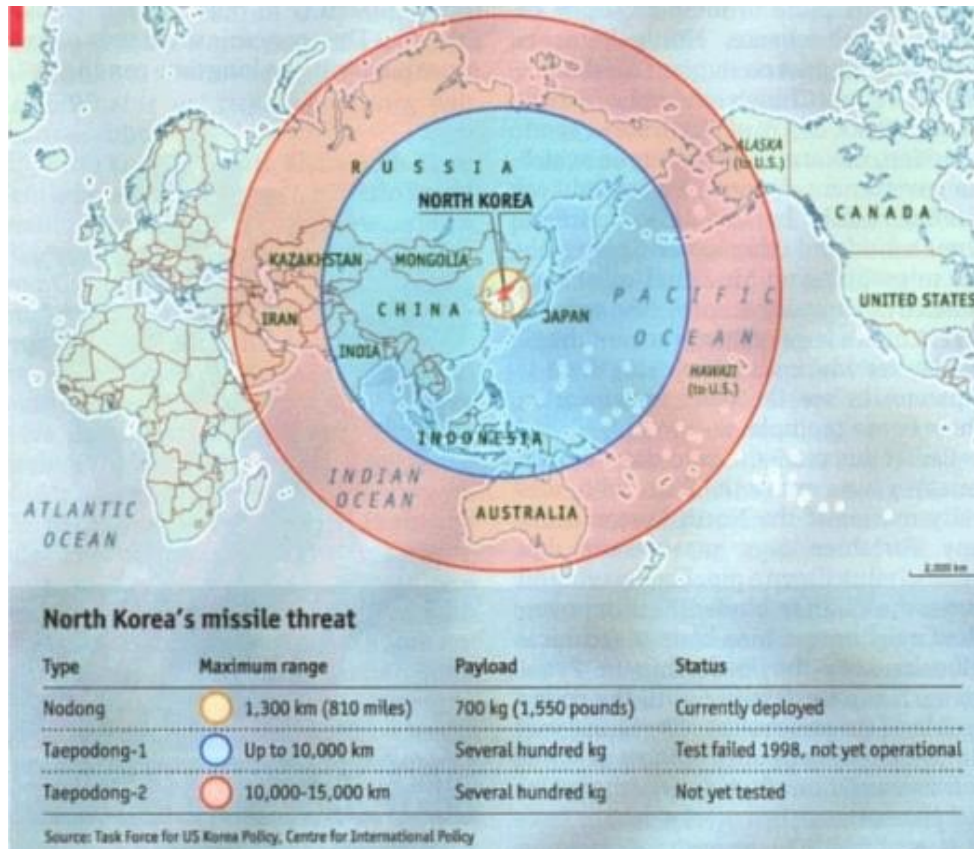


北韓長程飛彈的射程：
15,000 公里



連「經濟學人」都會犯錯的地圖誤用實例！

Maximum Range of North Korea's Missiles



The Economist

北韓長程飛彈
(15,000 公里)
居然未能涵蓋
美國本土！

地球圓週長：40,075 公里
地球半徑：6,371公里

2013, May 3rd Issue

2013, May 15th

Correction: North Korea's missiles

May 15th 2003 | From the print edition

[f Like](#) 0 [t Tweet](#) 4

Flat-earth thinking. Thank you to those readers who pointed out that, by superimposing concentric circles on a Mercator projection, the map in our May 3rd issue (now [corrected online](#)) greatly underestimated the potential reach of North Korea's missiles. We stand corrected.



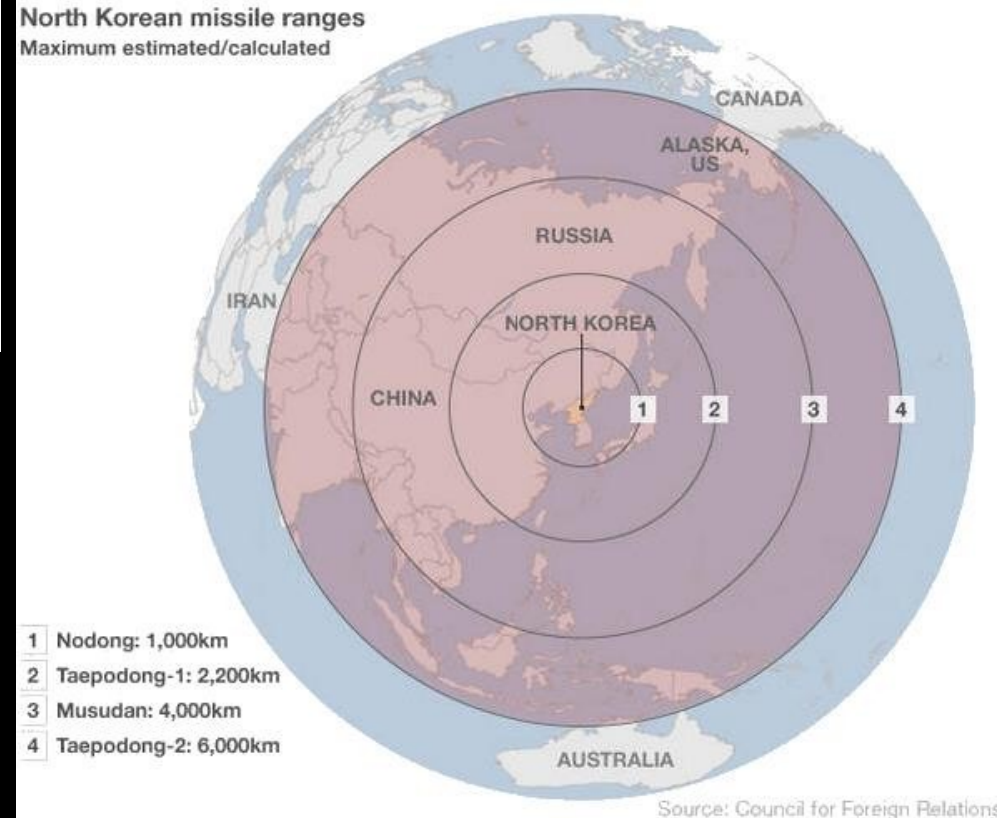
North Korea's missile threat

Type	Maximum range	Payload	Status
Nodong	1,300 km (810 miles)	700 kg (1,550 pounds)	Currently deployed
Taepodong-1	Up to 10,000 km	Several hundred kg	Test failed 1998, not yet operational
Taepodong-2	10,000-15,000 km	Several hundred kg	Not yet tested

Source: Task Force for US Korea Policy, Centre for International Policy

以「同心圓」的方式呈現飛彈射程範圍

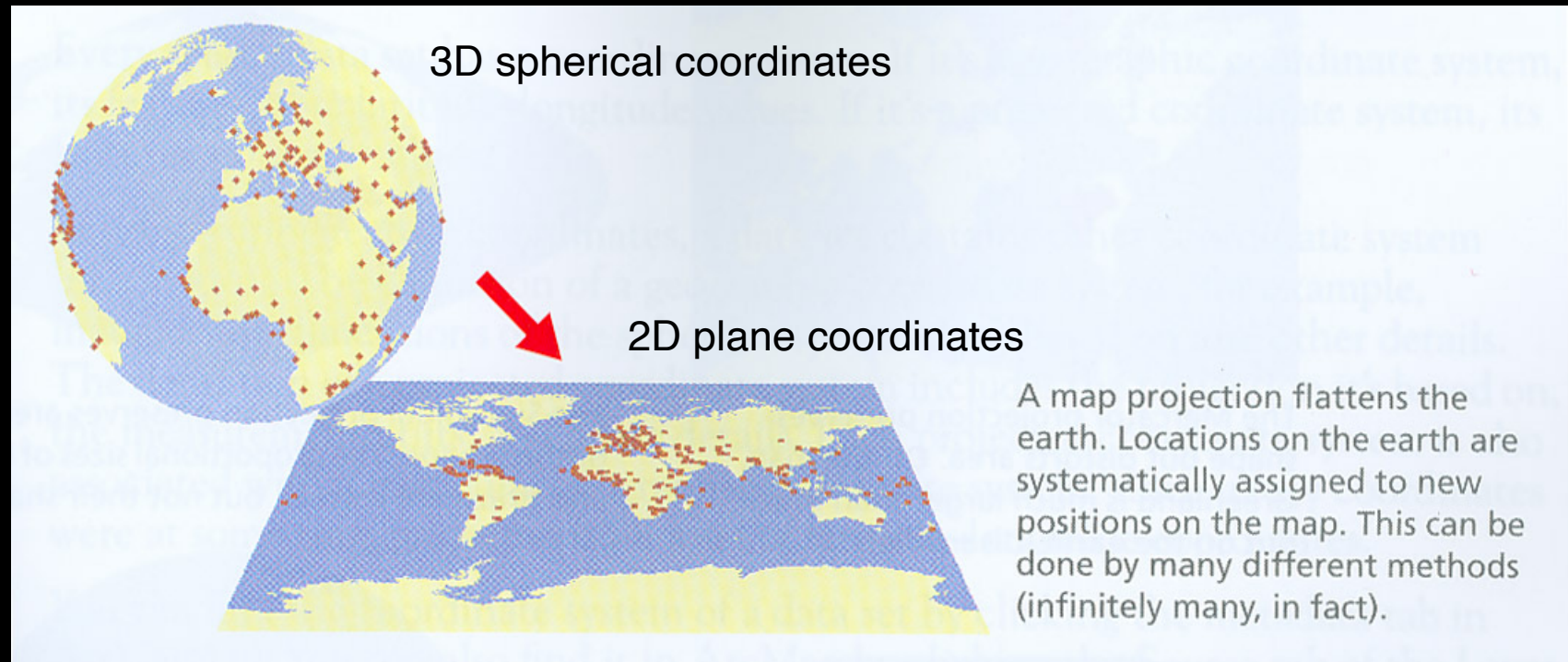
North Korean missile ranges
Maximum estimated/calculated



- 1 Nodong: 1,000km
- 2 Taepodong-1: 2,200km
- 3 Musudan: 4,000km
- 4 Taepodong-2: 6,000km

Source: Council for Foreign Relations

Map Projection: There Is No Perfect Projection!



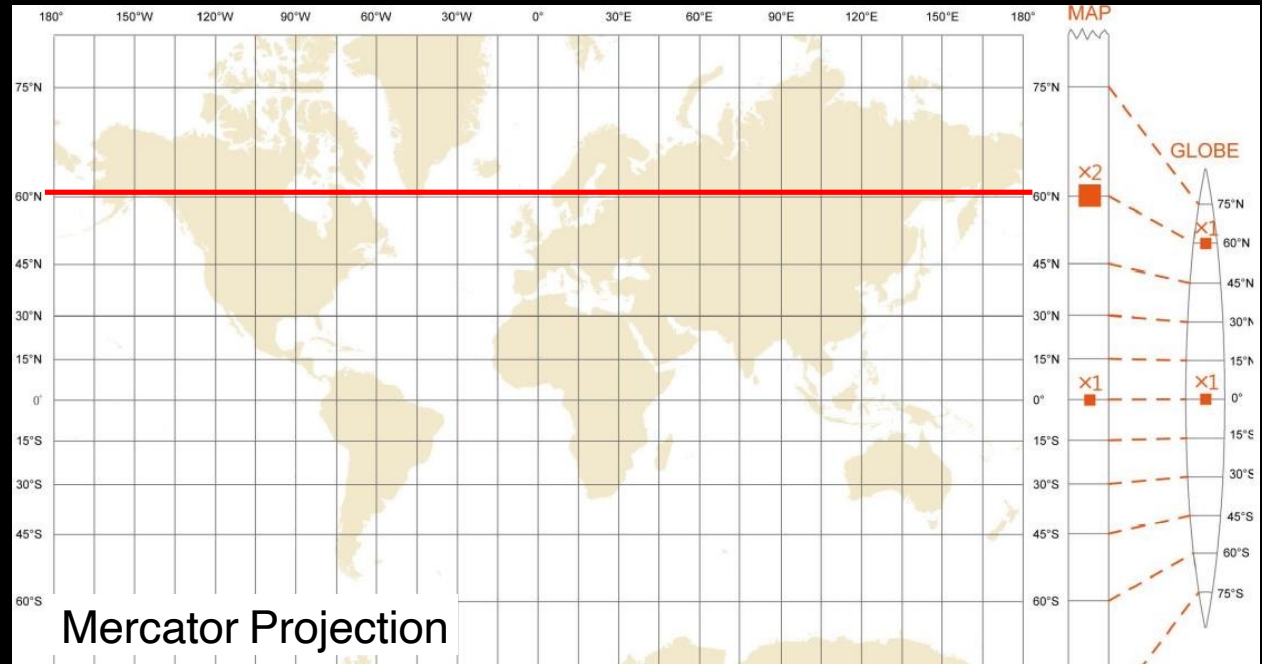
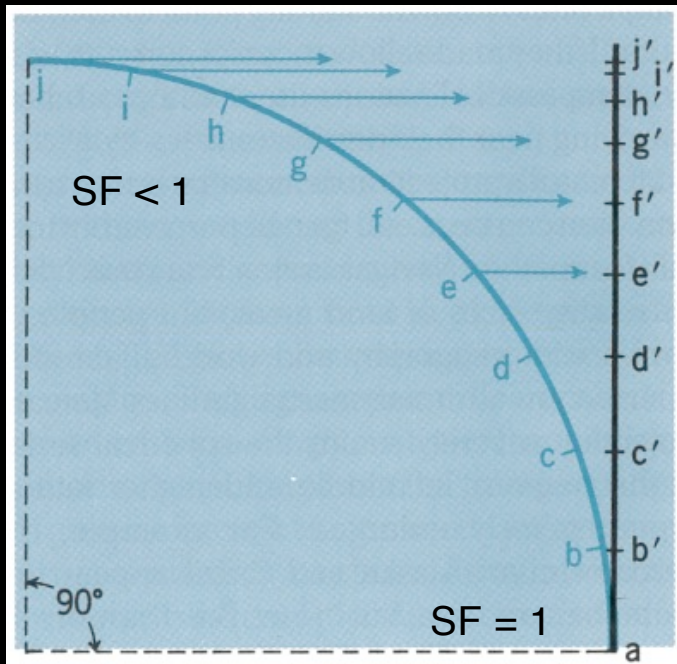
- Map projection is the orderly transfer of positions of places on the surface of the Earth to corresponding points on a flat sheet of paper, the map. (Frederick Pearson, 1990).

Scale Factor (SF) (尺度因子、縮尺係數)

- The lines on the projection can be compared to their corresponding lines on the reference globe by a ratio called scale factor (SF):

Actual scale

Principal scale



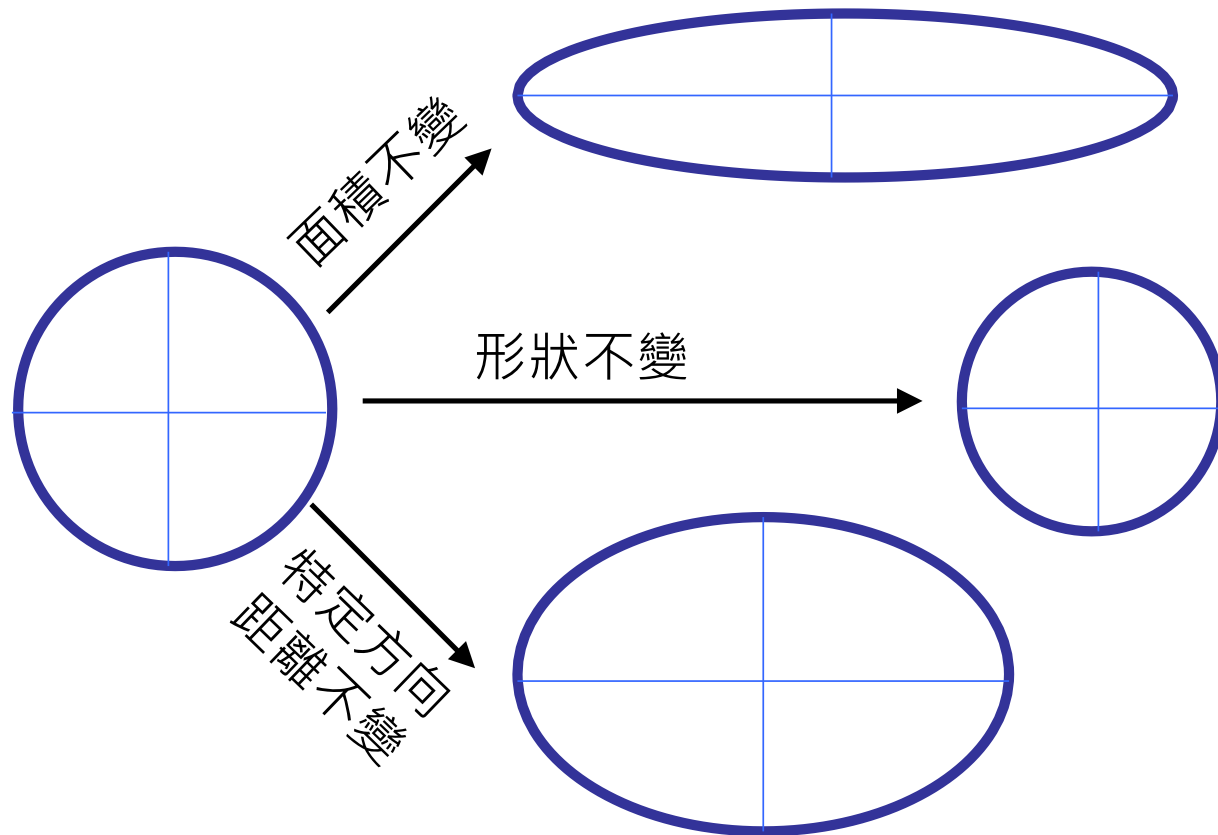
Elements of Map Projection

- Reference globe (參考地球儀)
 - Simplified model of the Earth, which served as the object being projected
- Standard line and standard point (標準線 / 標準點)
 - Locations where distortion is minimized (true scale)
- Developed surface (投影面)
 - Geometric surface onto which the Earth is projected
- Light source (光源)
 - Determining how features are transferred to the map
- Aspect (投影方位)
 - Orientation of the projection surface
- Spatial properties

Spatial Properties of Map Projection

- Conformal (or orthomorphic) projection (正形投影) **Shape**
 - Meridians intersect parallels at right angles, and the scale is the same in all directions
- Equal area (or equivalent) projection (等積投影) **Area**
 - Important for thematic maps at a global scale (when area matters, e.g. population, land use, etc.)
- Equidistant projection (等距投影) **Distance**
 - Preservation of great circle distances
- Azimuthal (or true-direction) projection (正方位投影) **Direction**
 - True directions are shown from one central point to all other points
- Minimum error (compromise) projection (折衷投影)
 - Hybrid projection attempts to control or minimize all four map projection properties.

Spatial Properties of Map Projection

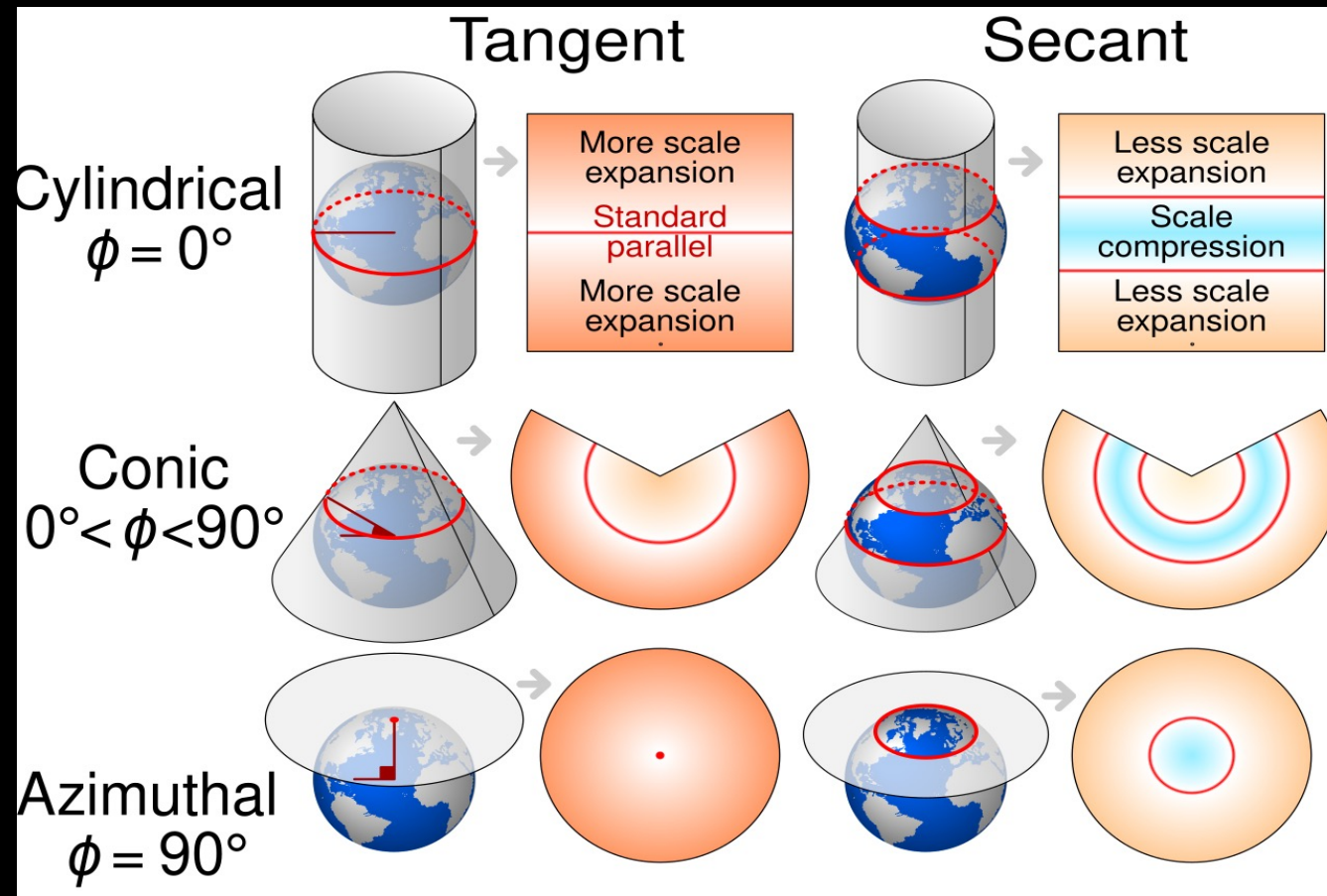


Elements of Map Projection

- Reference globe (參考地球儀)
 - Simplified model of the Earth, which served as the object being projected
- Standard line and standard point (標準線 / 標準點)
 - Locations where distortion is minimized (true scale)
- Developed surface (投影面)
 - Geometric surface onto which the Earth is projected
- Light source (光源)
 - Determining how features are transferred to the map
- Aspect (投影方位)
 - Orientation of the projection surface
- Spatial properties

Standard Line and Standard Point (標準線 / 標準點)

- Tangent (切投影) and secant (割投影)

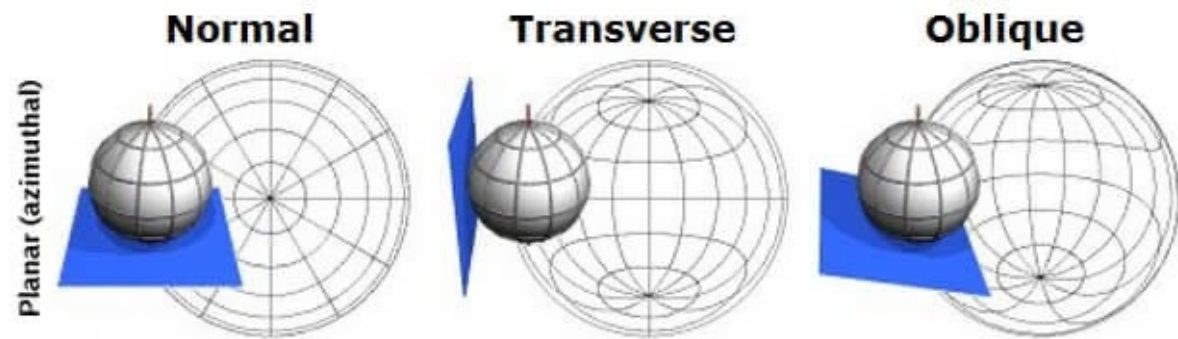


Aspect (refer to the developed surface)

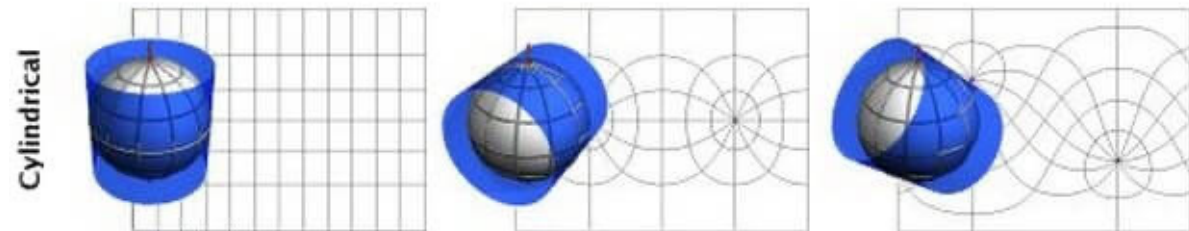
Developed surface

1. 平面 (方位投影)

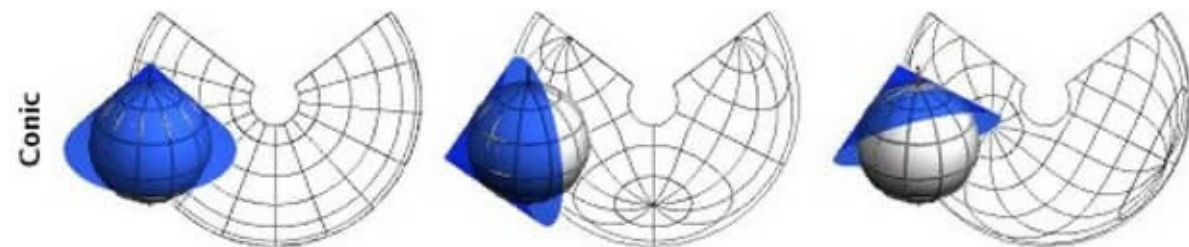
Light source variation



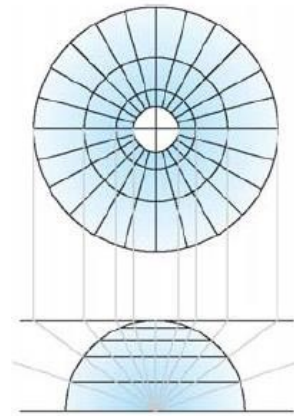
2. 圓柱投影



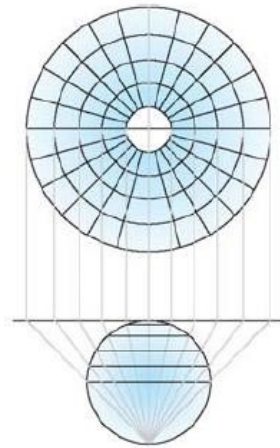
3. 圓錐投影



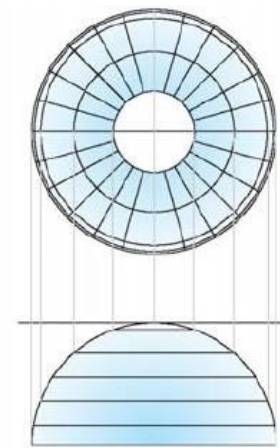
Light source



Gnomonic
1-1. 心射



Stereographic
1-2. 平射
(正形投影)



Orthographic
1-3. 正射

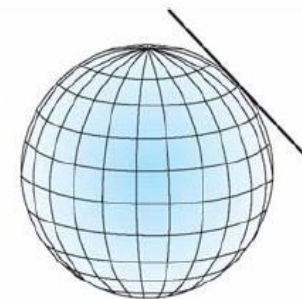
Aspect (refer to the globe)



Polar



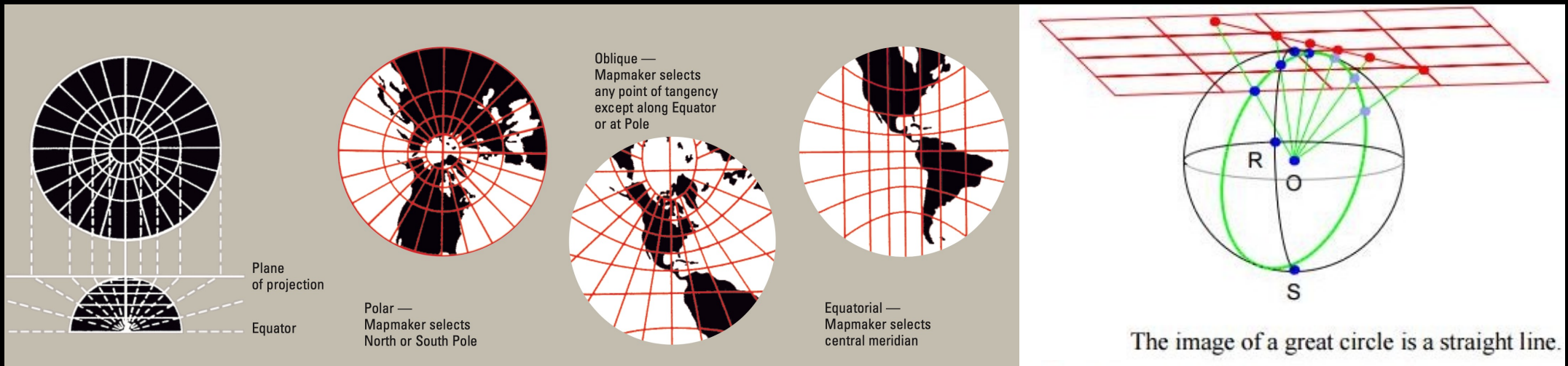
Equatorial



Oblique

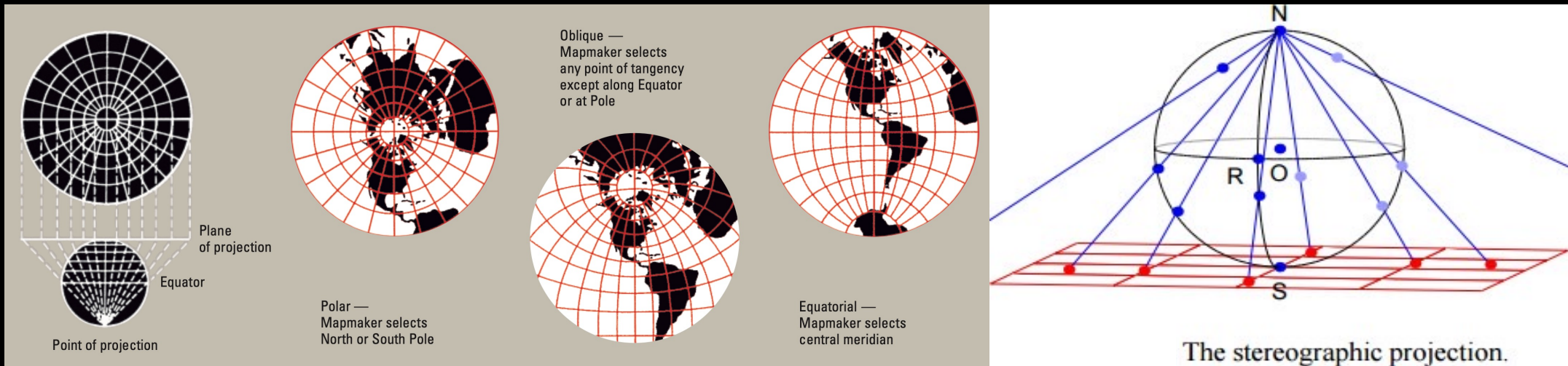
1-1. Gnomonic Projection (心射投影)

- Any straight line drawn on the map is on a great circle.
- **Directions** are true *only* from center point of projection.
- **Scale** and **distortion** increase vary rapidly away from center point.
- Map is perspective (透視), but not conformal, equal area, or equidistant.
- Used by navigators to find the shortest path between two points, e.g. mapping seismic waves



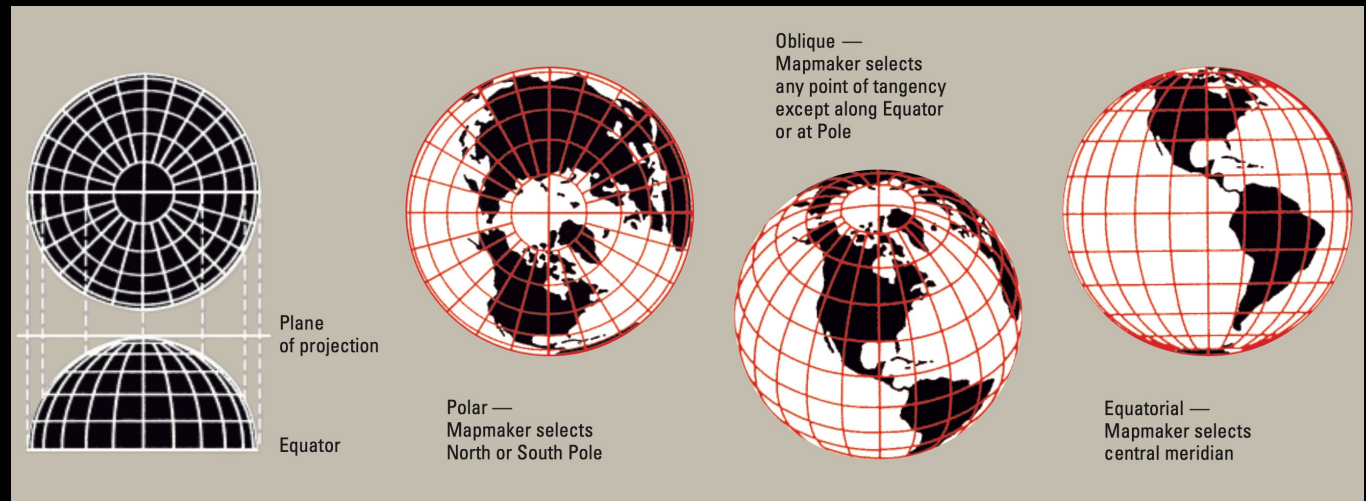
1-2. Stereographic Projection (平射投影)

- Any straight line through center point is a great circle.
- **Directions** are true *only* from center point of projection.
- **Scale** and **distortion** increase away from center point.
- Map is perspective and conformal, but not equal area or equidistant.
- Used to map large continent-size areas, e.g. polar aspects for topographic maps and charts for navigating in latitudes above 80°



1-3. Orthographic Projection (正射投影)

- Any straight line through center point is a great circle.
- **Directions** are true *only* from center point of projection.
- **Scale** decreases along all lines radiating from center point of projection.
- **Distortion** of shapes and areas increase away from center point.
- Map is perspective, but not conformal or equal area. In the polar aspect, distances are true along the Equator and all other parallels.
- Used for perspective views of the Earth and other planets
- The Earth appears as it would on a photograph from deep space.

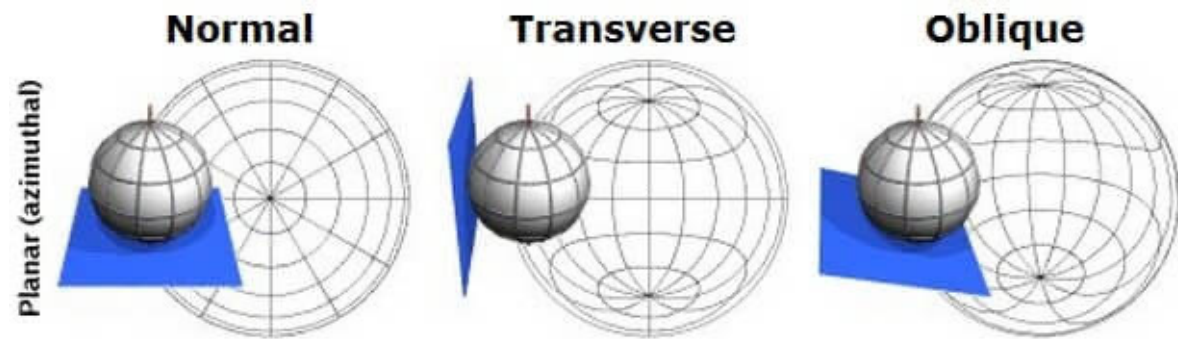


Aspect (refer to the developed surface)

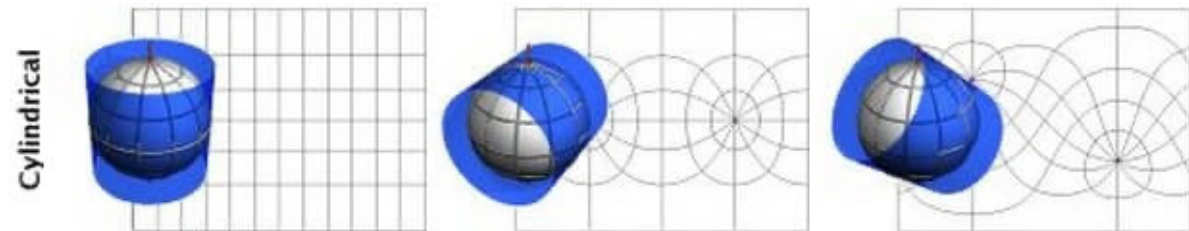
Developed surface

1. 平面 (方位投影)

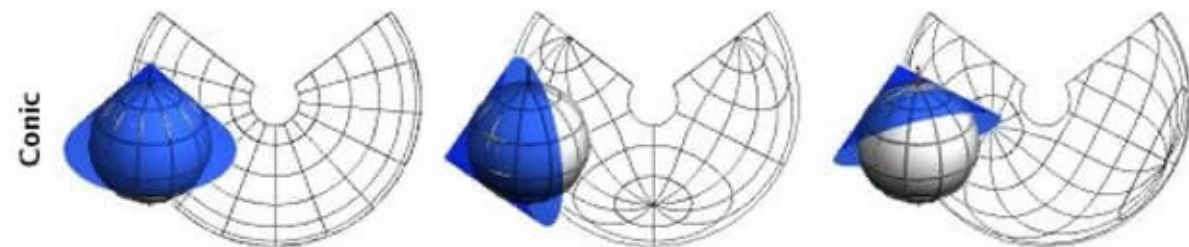
Light source variation



2. 圓柱投影

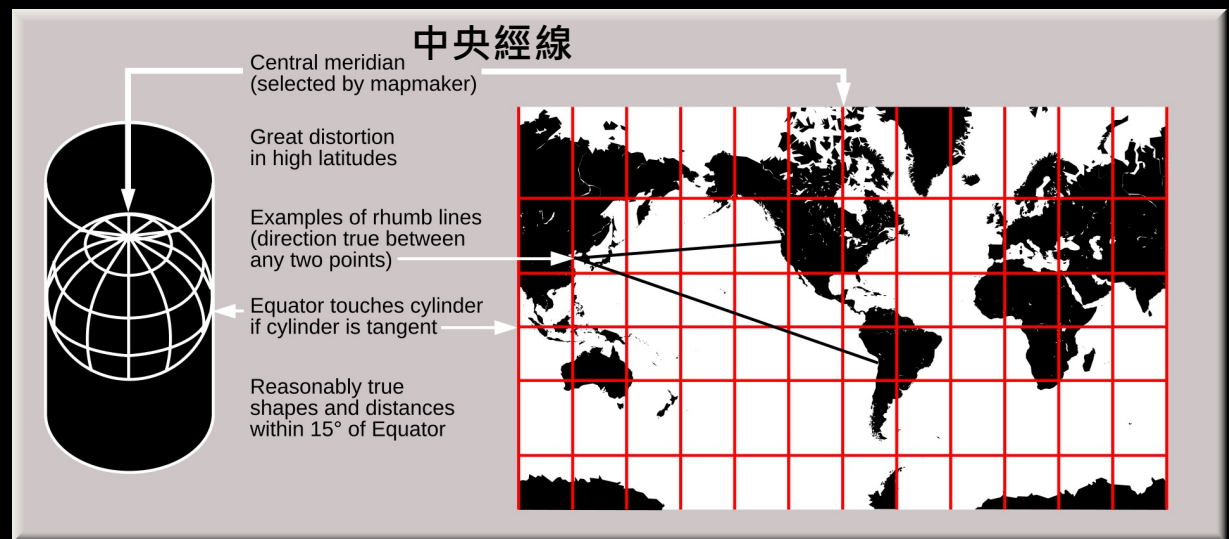


3. 圓錐投影



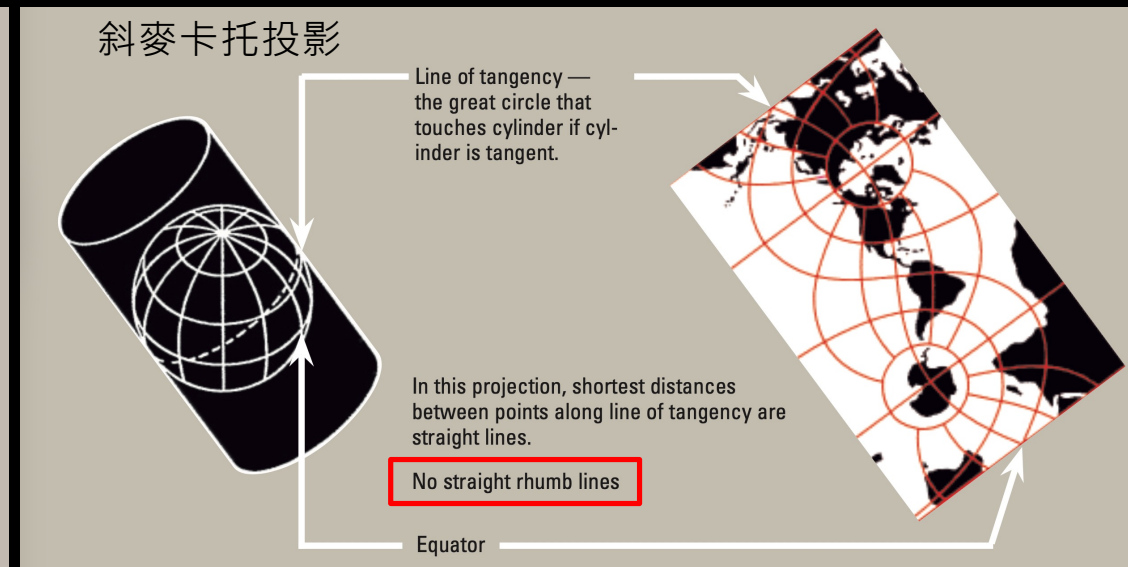
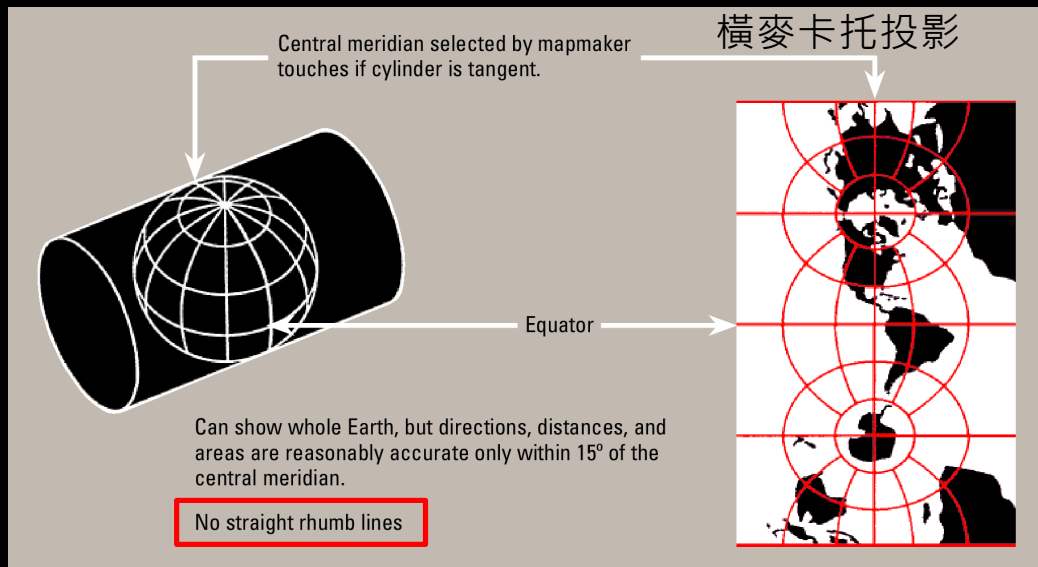
2. Cylindrical Projection (圓柱投影)

- Mercator (麥卡托投影): Equatorial aspect
- In the tangent case, the equator is the standard parallel. The standard line is also a great circle.
- In the secant case, the two standard parallels (small circles) will be located north and south of the equator.
- **Distance** are true *only* along Equator.
- **Directions** along a rhumb line are true between *any* two points, but a rhumb line is usually *not* the shortest distance between points.
- Rhumb line is a straight line.



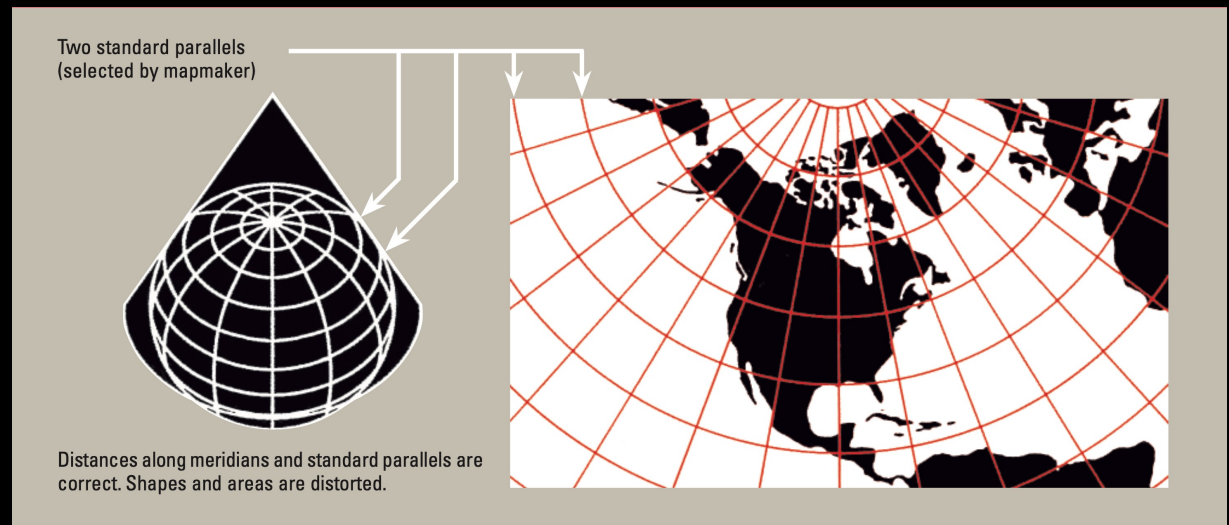
2. Cylindrical Projection (圓柱投影)

- Transverse Mercator (橫麥卡托投影)
- **Distances** are true *only* along the central meridian selected by the mapmaker.
- The map is conformal, and **shapes and angles** within any small area are essentially true.



3. Conic Projection (圓錐投影)

- Simple conic (equidistant conic) (簡單圓錐投影 / 等距圓錐投影)
- **Distances** are true *only* along all meridians and along one or two standard parallels.
- **Directions, shapes, and areas** are reasonably accurate, but **distortion** increases away from standard parallels.
- Map is not conformal, perspective, or equal area.



Spatial Properties of Map Projection

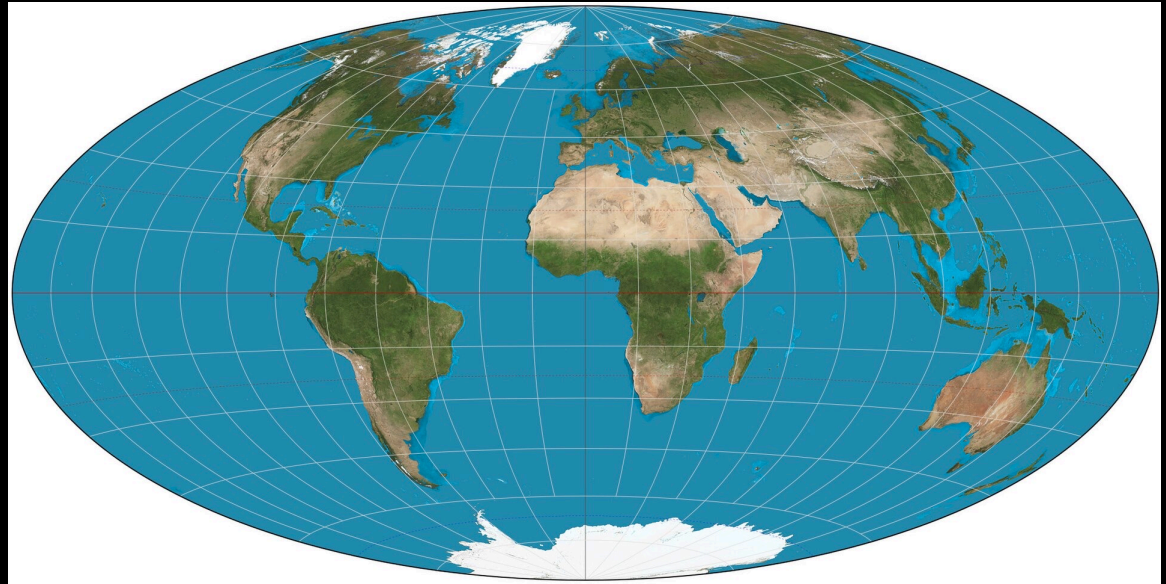
- Conformal (or orthomorphic) projection (正形投影) **Shape**
 - Meridians intersect parallels at right angles, and the scale is the same in all directions
- Equal area (or equivalent) projection (等積投影) **Area**
 - Important for thematic maps at a global scale (when area matters, e.g. population, land use, etc.)
- Equidistant projection (等距投影) **Distance**
 - Preservation of great circle distances
- Azimuthal (or true-direction) projection (正方位投影) **Direction**
 - True directions are shown from one central point to all other points
- Minimum error (compromise) projection (折衷投影)
 - Hybrid projection attempts to control or minimize all four map projection properties.

Trade Off between Area and Shape

Mercator Projection (Conformal)

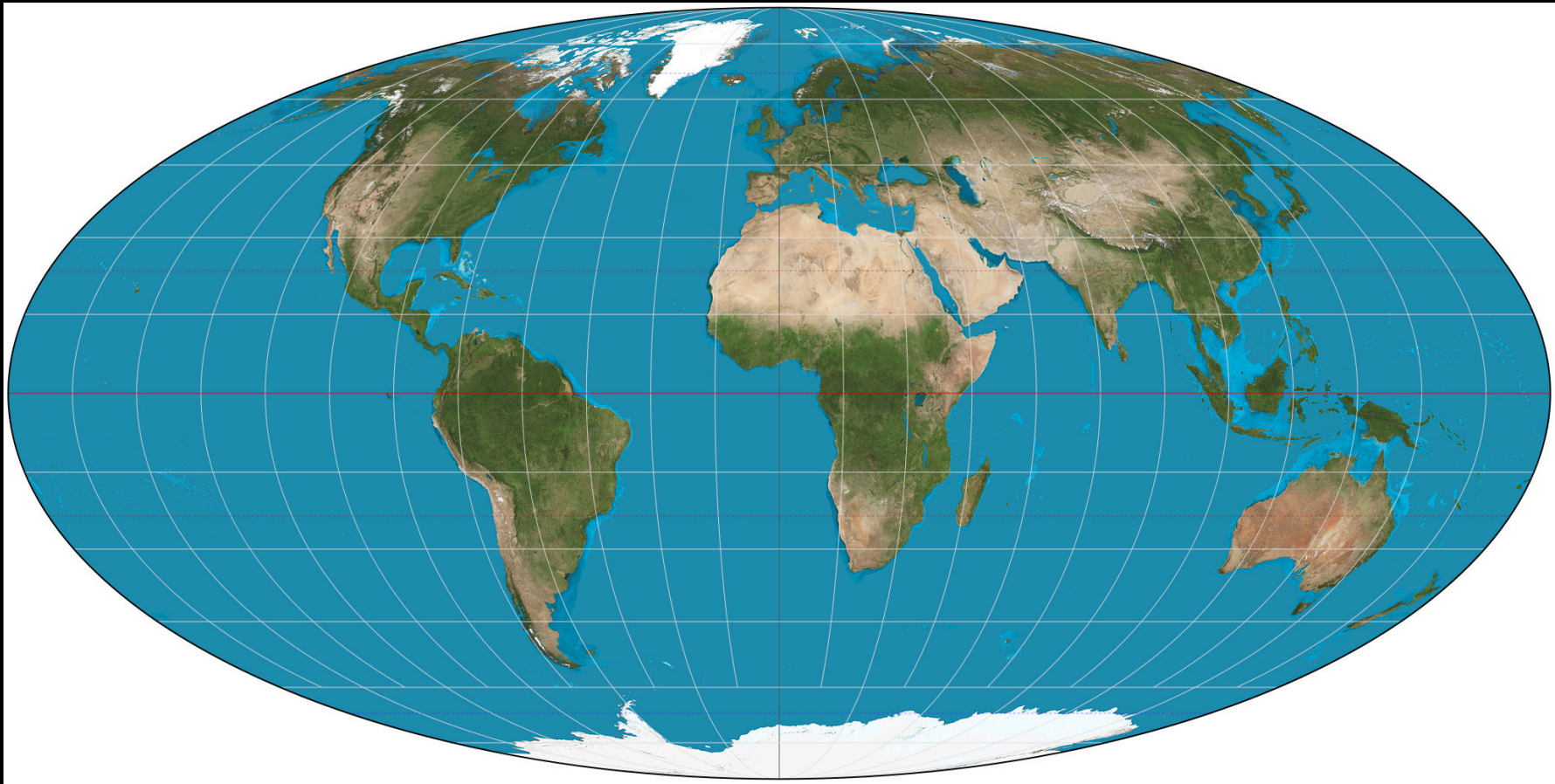


Hammer Projection
(Equal-area, modified azimuthal)

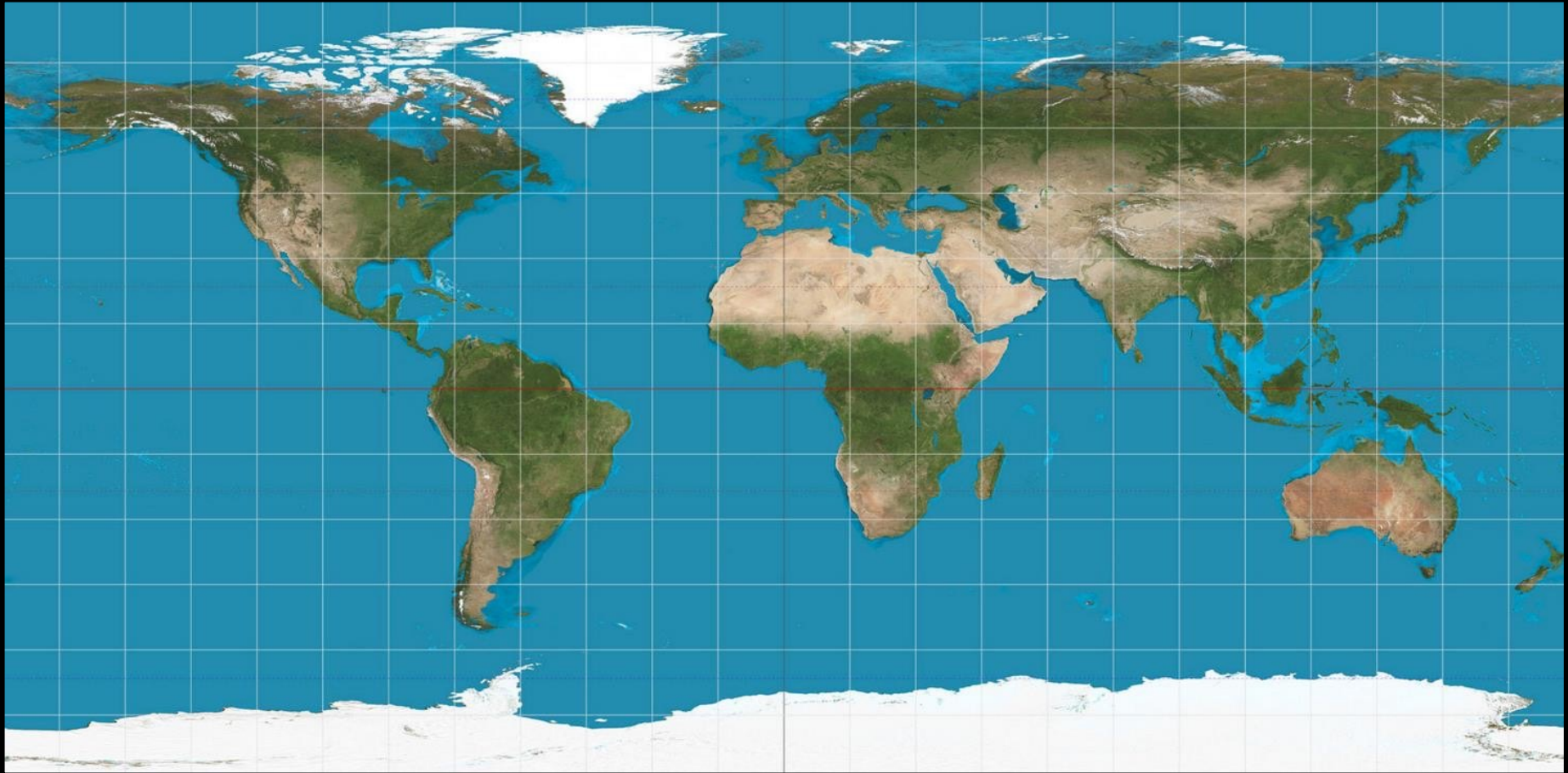


Mollweide Projection (Equal-Area)

Pseudocylindrical (偽圓柱投影)

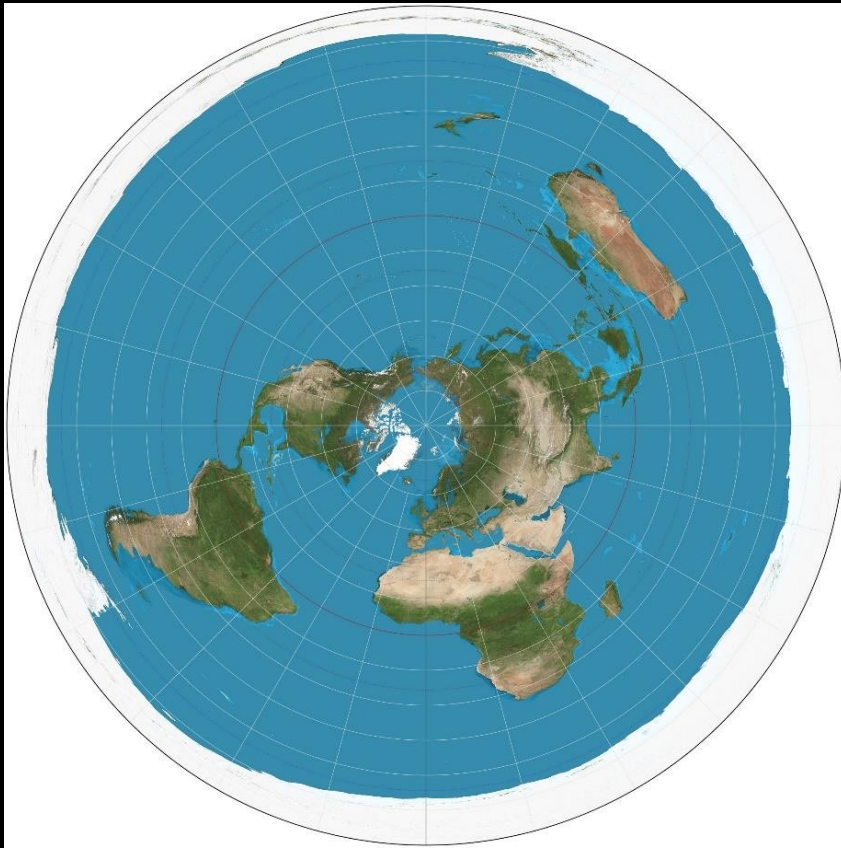


Equirectangular (Cylindrical + Equidistant) (方格投影)

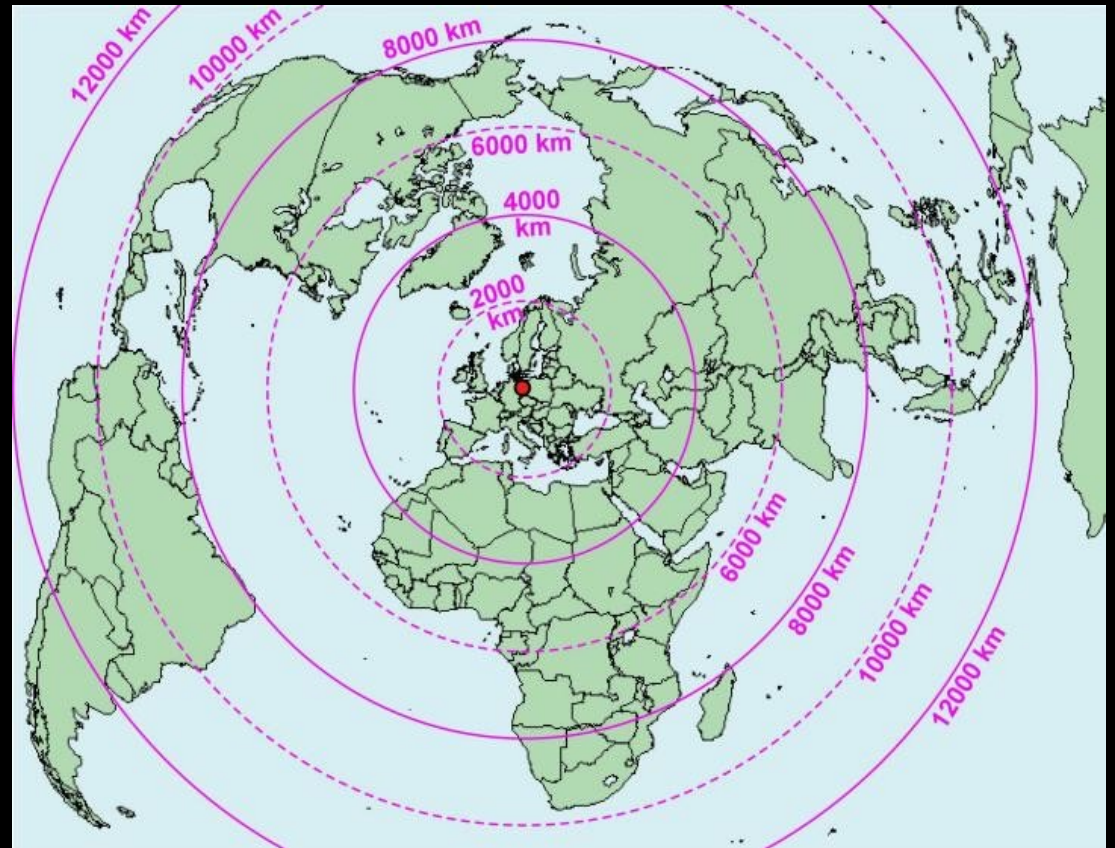


Azimuthal Equidistant Projection (方位等距投影)

Centered at North Pole



Centered at Berlin



Minimum Error (or Compromise) Projection

The New York Times

The Impossible Quest for the Perfect Map

By John Noble Wilford

Oct. 25, 1988

This month, the National Geographic Society announced it is changing the projection for its world maps, beginning with the one to be distributed in the December issue of the National Geographic magazine. The Van der Grinten projection, in use since 1922, will be replaced by one developed by Arthur H. Robinson, one of the nation's most respected cartographers.



The Robinson map projection is shown centered on Greenwich.

FAST FACT

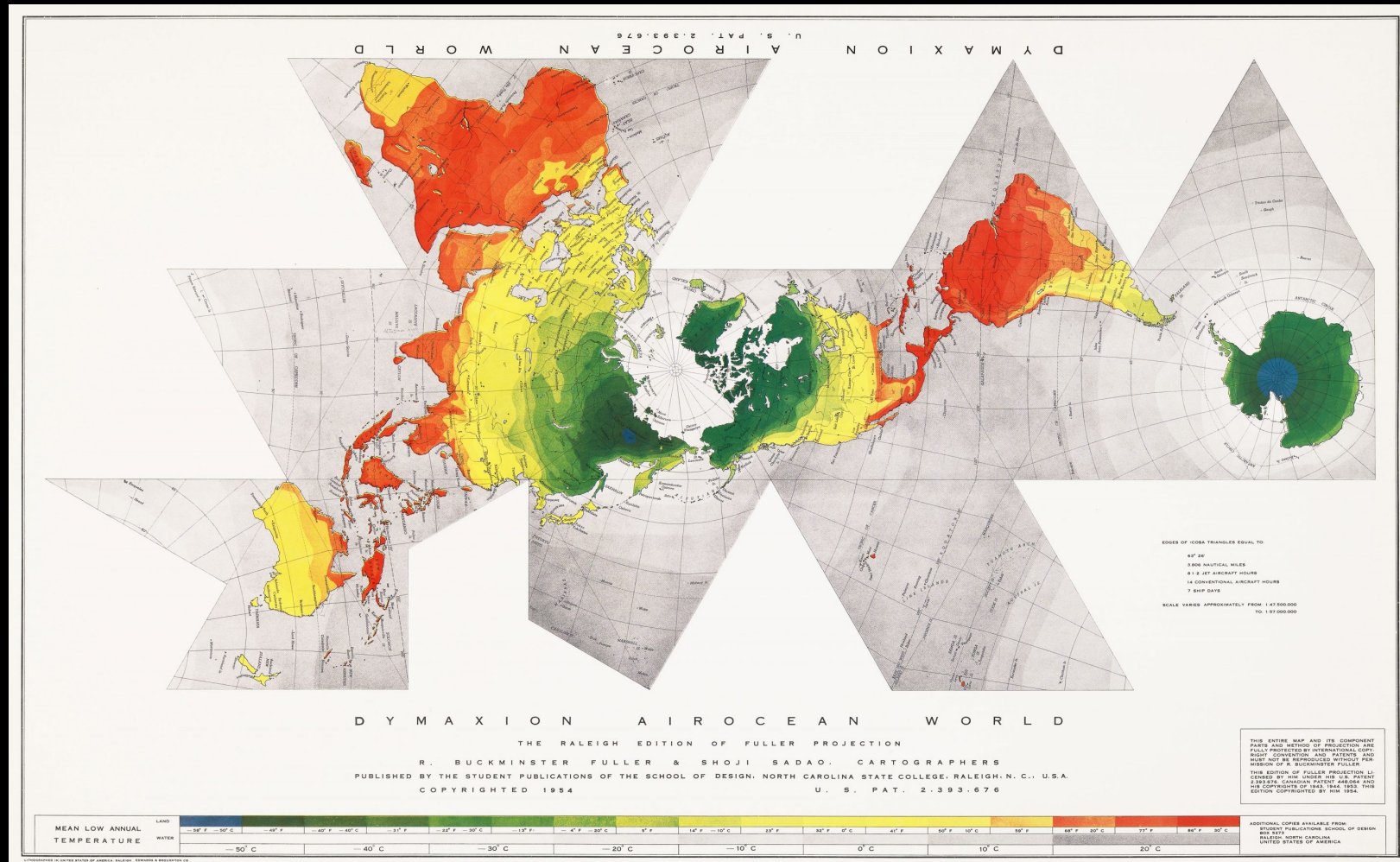


In 1995, the Winkel Tripel projection replaced the Robinson projection on the Society's signature world maps. Long used in various European atlases, the Winkel Tripel, first published as a map supplement in *National Geographic Magazine* in April 1995, is one of the most accurate representations of the round globe on flat paper.



The Winkel Tripel projection is shown centered on Greenwich.

Dymaxion Map



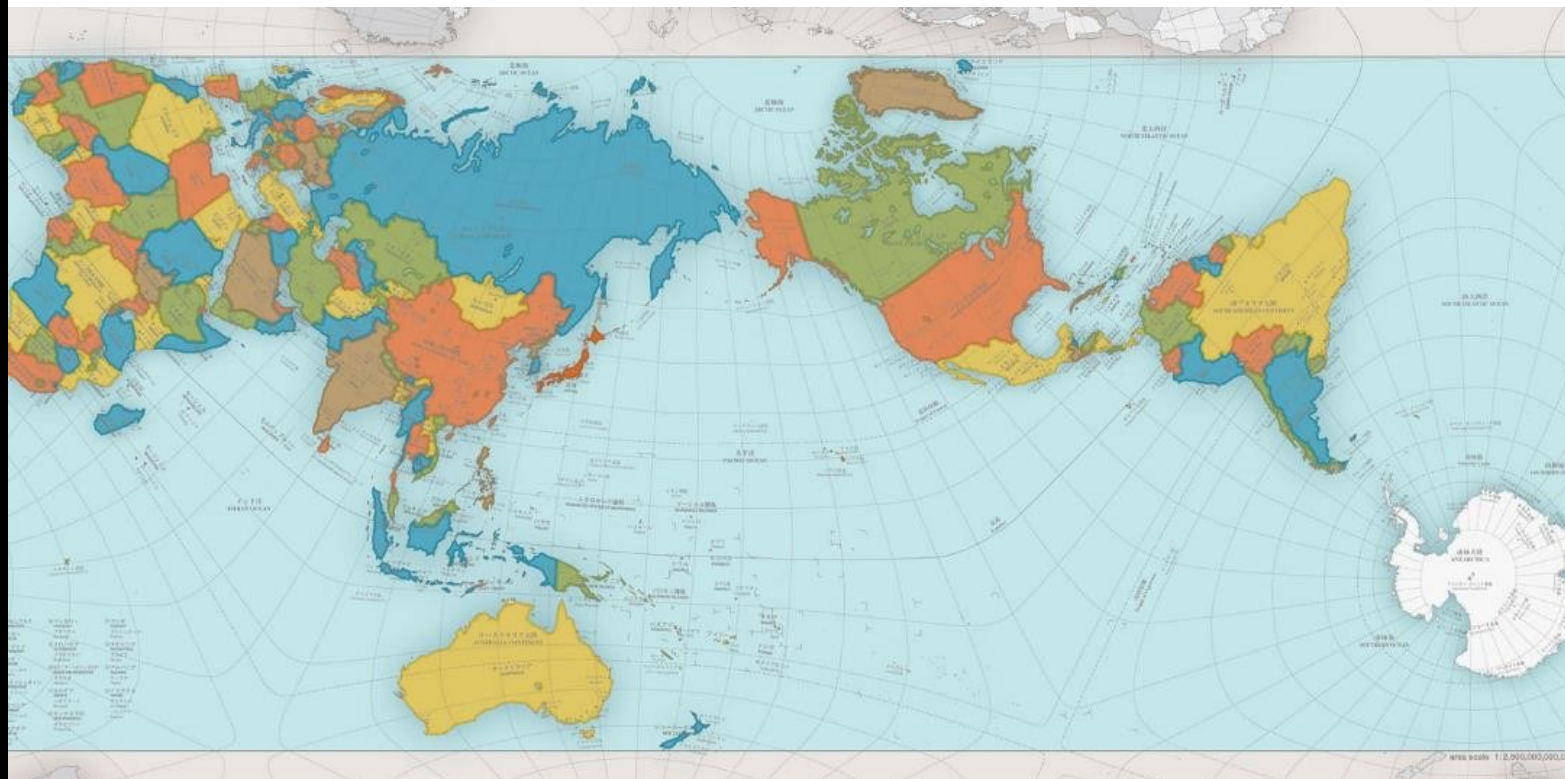
Dymaxion Map, 1943, Buckminster Fuller

AuthaGraph World Map

2016

Good Design Award 2016 results. Each Special Awards were selected from [GOOD DESIGN BEST 100] and announce at October 28, first day of GOOD DESIGN EXHIBITION 2016.

► Outline of 2016 award winners



Summary

● = Yes
○ = Partly

Properties

Suitable for Mapping

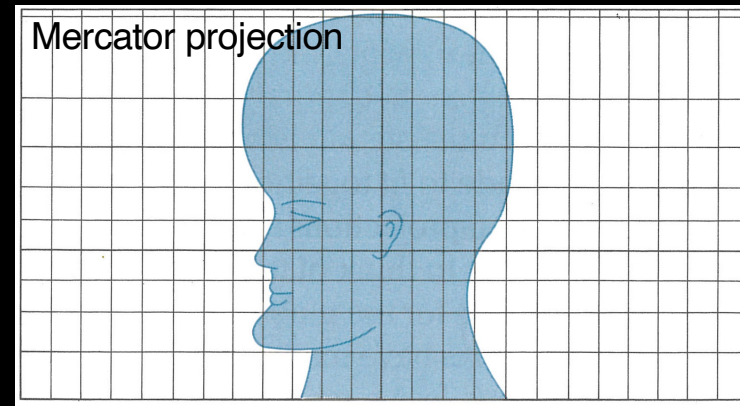
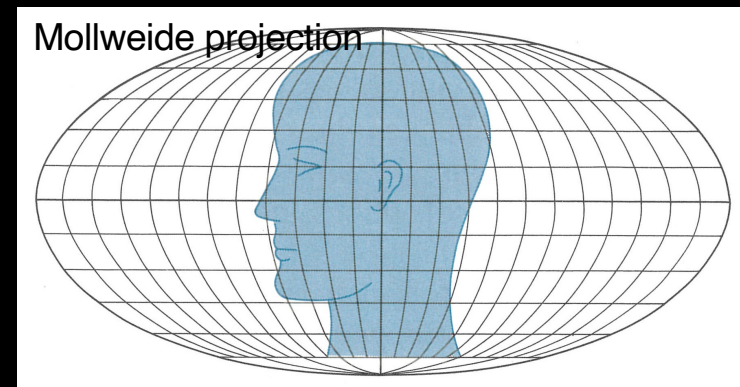
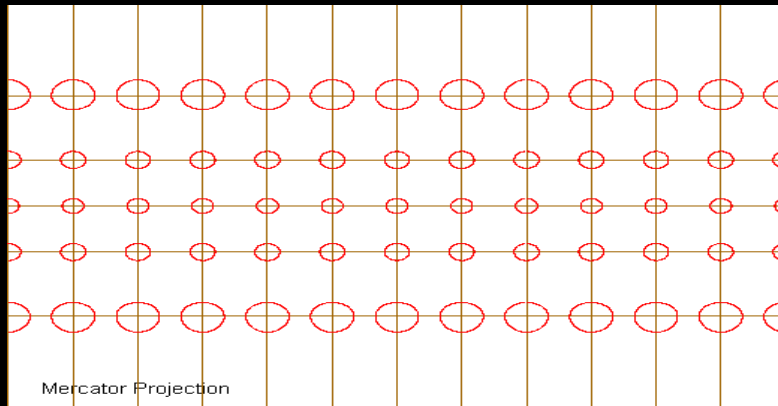
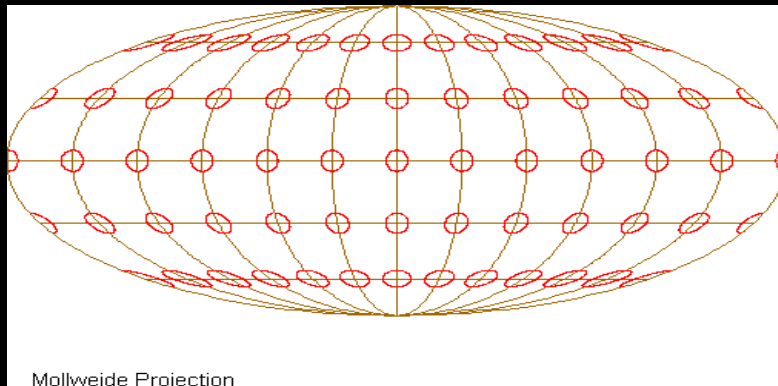
General Use

Projection	Type	Conformal						World					Topographic Maps				
		Equal Area	Equidistant	True Direction	Perspective	Compromise	Straight Rhumbs	Hemisphere	Continent/Ocean	Region/Sea	Medium Scale	Large Scale	Geological Maps	Thematic Maps	Presentations	Navigation	USGS Maps
Globe	Sphere	●	●	●	●			●						●	●		
Mercator	Cylindrical	●		○		●		○		●			●	●		●	●
Transverse Mercator	Cylindrical	●							●	●	●	●	●	●			●
Oblique Mercator	Cylindrical	●							●	●	●	●	●				●
Space Oblique Mercator	Cylindrical	●									●		●				●
Miller Cylindrical	Cylindrical				●			●						●			●
Robinson	Pseudocylindrical				●			●						●	●		
Sinusoidal Equal Area	Pseudocylindrical		●	○				●	●					●			●
Orthographic	Azimuthal			○	●				○								●
Stereographic	Azimuthal	●		○	●			●	●	●	●	●	●	●		●	●
Gnomonic	Azimuthal			○	●					○					●	●	
Azimuthal Equidistant	Azimuthal		○	○				○	●	●	●	○	●				●
Lambert Azimuthal Equal Area	Azimuthal		●	○				●	●	●				●	●		●
Albers Equal Area Conic	Conic		●						●	●	●			●	●		●
Lambert Conformal Conic	Conic	●		○					●	●	●	●	●	●		●	●
Equidistant Conic (Simple Conic)	Conic		○						●	●							
Polyconic	Conic		○		●						○	○	●				●
Bipolar Oblique Conic Conformal	Conic	●							●					●			●

All above projections (except Robinson) are explained in detail in Map Projections — A Working Manual, John P. Snyder, U. S. Geological Survey, Professional Paper 1395 (Washington: USGPO, 1987, 383 pp.)

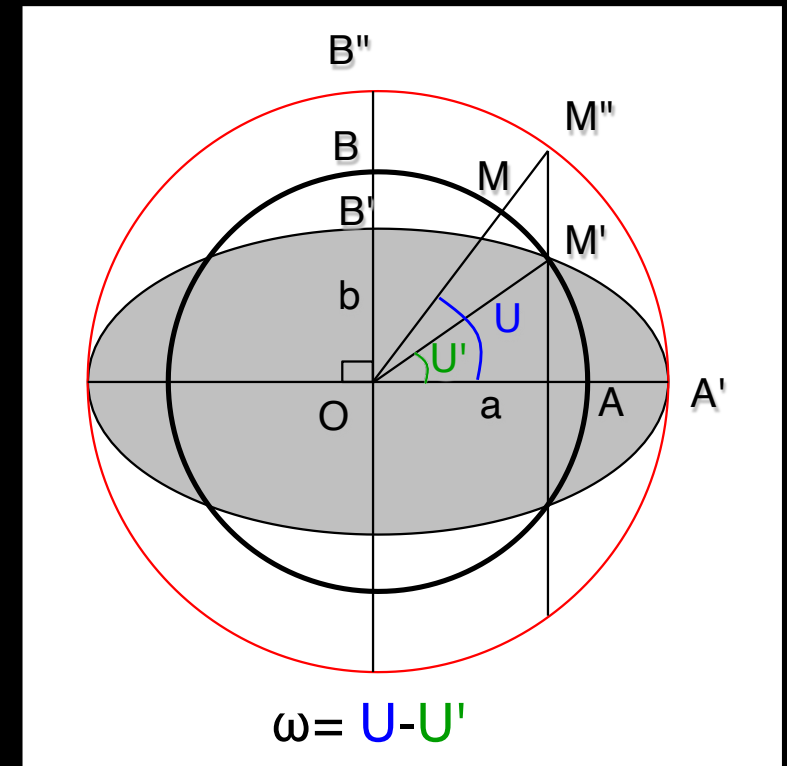
Determining the Distortion

- Depicting a geometrical figure (square or circle) or familiar object (human head) and plot is at several locations on the projection graticule.

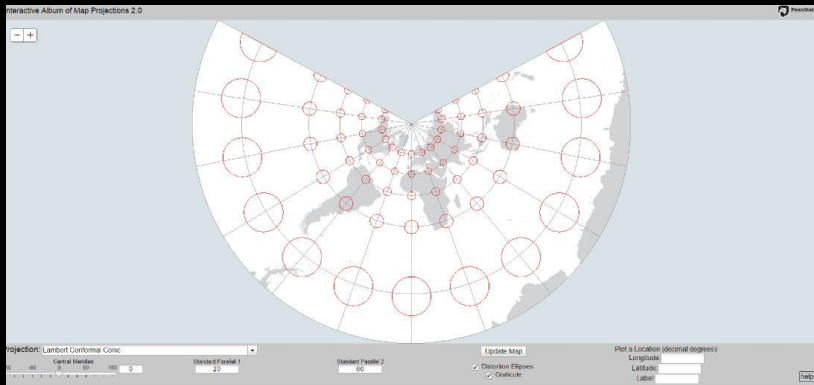


Determining the Distortion

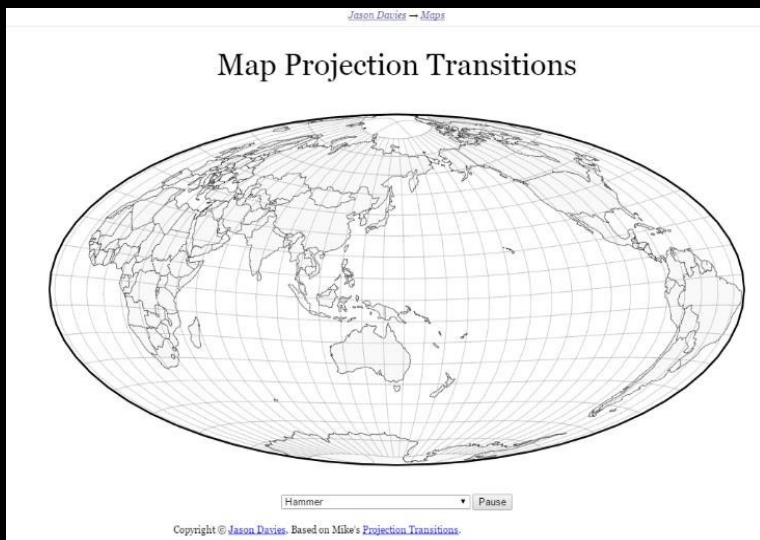
- Tissot's indicatrix (底索指示線) (法國數學家, Nicolas Auguste Tissot 提出)
- Ellipse of distortion (變形橢圓)
 - a: semi-major axis (半長軸)
 - b: semi-minor axis (半短軸)
 - ω : $U-U'$ (原本角度－結果角度)
- a b product & ω to represent distortion
- Area change: $S = \pi ab$
 - On Equator, $OA=OB=1.0$ ($a=b$), ω is unchanged
 - Grey ellipse (equal-area)
 - $ab = 1.25 \times 0.8 = 1.0$
 - Red circle (conformal)
 - $ab = 1.25 \times 1.25 = 1.5625$



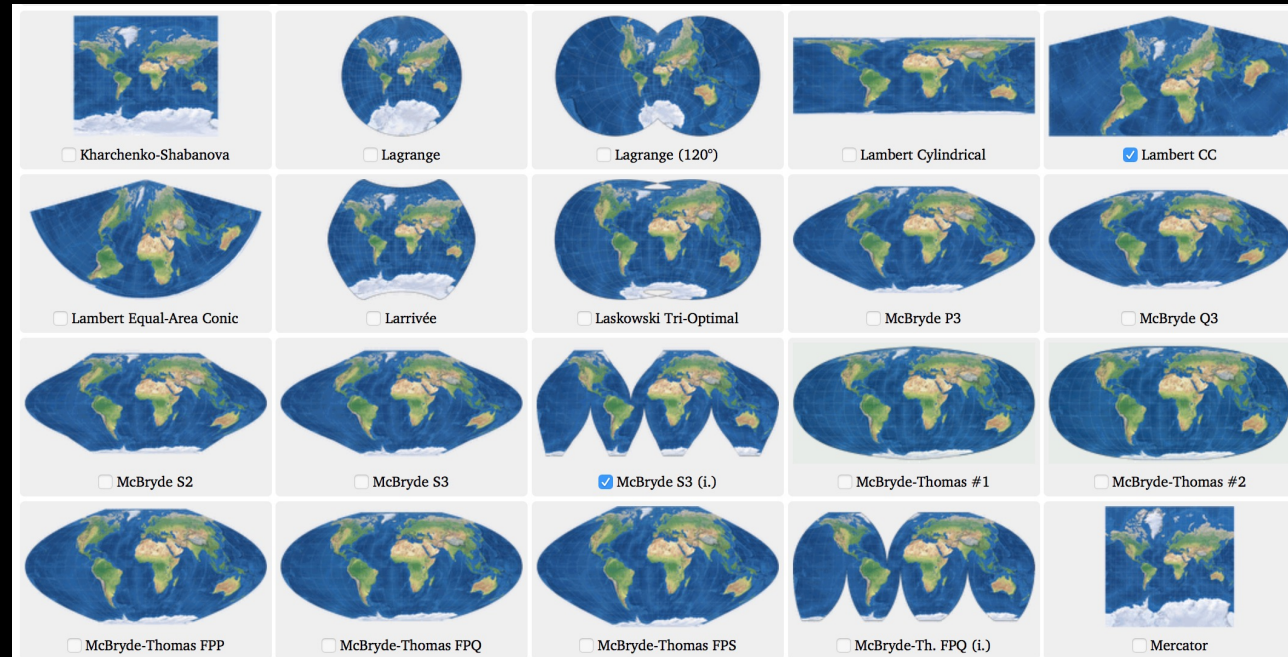
Play with Interactive Map Projections!



<https://projections.mgis.psu.edu>



<https://www.jasondavies.com/maps/transition/>



<https://map-projections.net/imglist.php>

Homework 2 – Due 2PM, March 27

- A pdf file includes 5 maps + 2-page report.
- Choose one the interactive map projection website (last slide)
- Choose one map projection for each spatial property type (conformal, equal area, equidistant, azimuthal, and compromise) (5 in total)
- Describe each map projection (name, spatial property, direction, shape, area, distortion, etc.)
- Choose one geographic topic you learned from the class of physical or human geography (or in your daily life), and explain which one of the five map projections is the best projection for this topic.

Primary Coordinate System in Taiwan

- Geographic coordinate system (GCS)
 - WGS84 – Latitude / Longitude
 - TWD97 – Latitude / Longitude
- Projected coordinate system (PCS)
 - Universal Transverse Mercator (UTM)
 - TW67 – TM2
 - TW97 – TM2

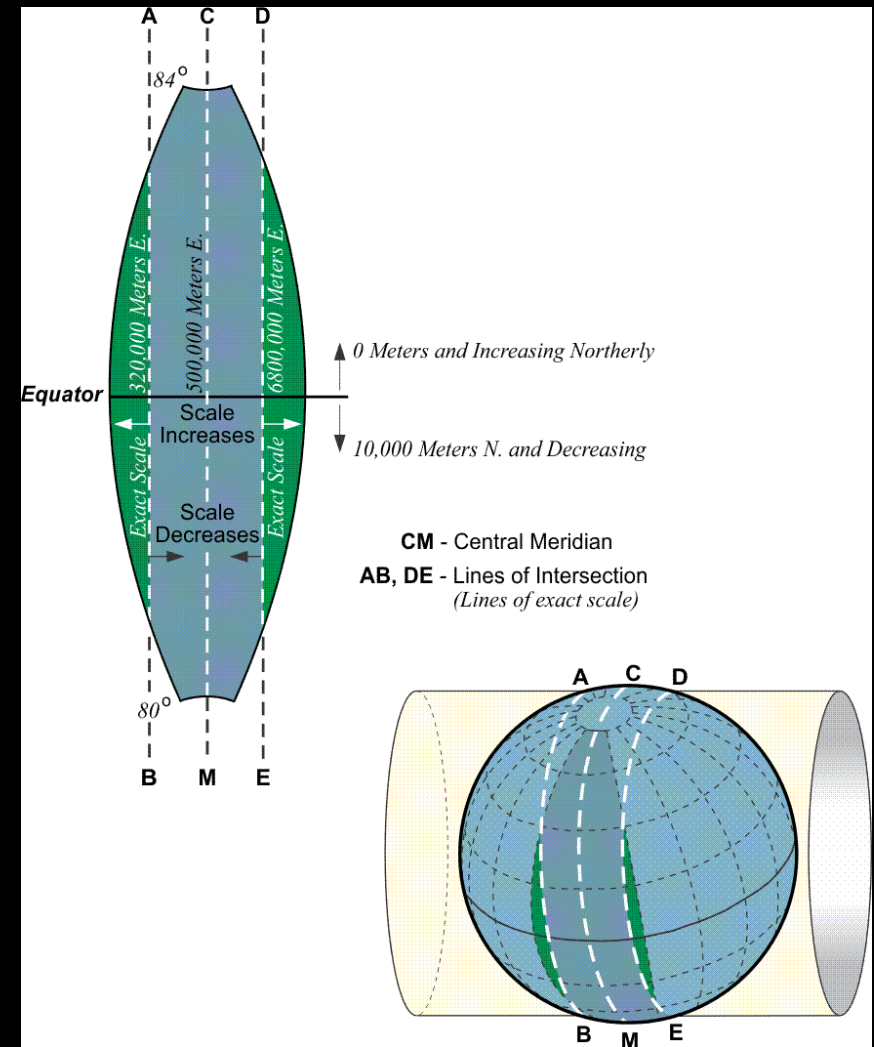
完整的投影坐標系統命名：Datum + Projection，例如：TWD 97- TM2

Universal Transverse Mercator (UTM) (通用橫麥卡托投影)

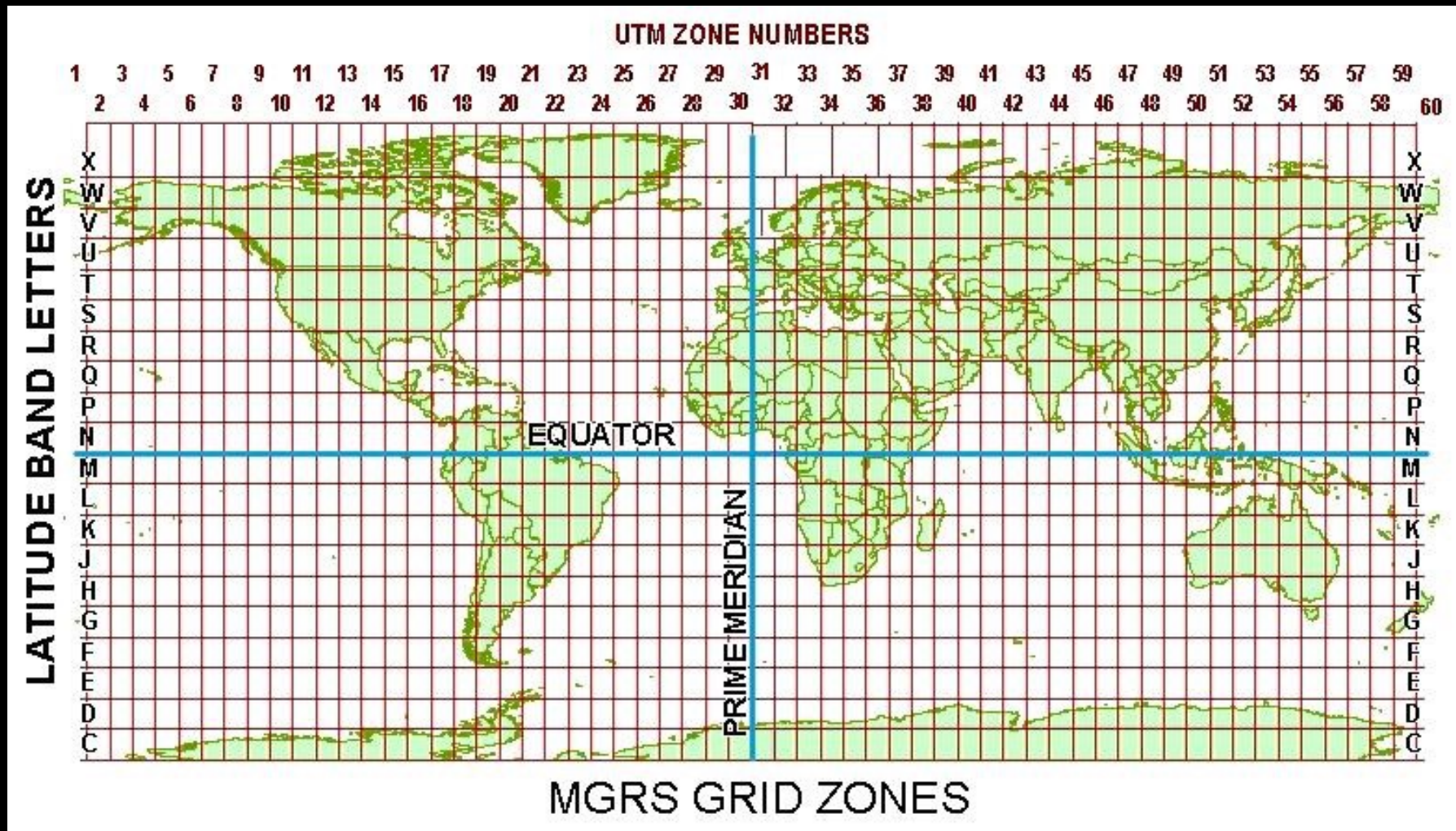
- Based on transverse cylindrical projection projecting the sphere onto a cylinder tangent to a central meridian.
- The central meridian (and area of least distortion) can be moved around the earth by rotating the cylinder.
- UTM zone numbers designate 6-degree longitudinal strips extending from 80 degrees South latitude to 84 degrees North latitude.
- UTM zone characters designate 8-degree zones extending north and south from the equator.

Universal Transverse Mercator (UTM) (通用橫麥卡托投影)

- Units: meters
- False easting
 - from the central meridian
 - with a 500km false easting to insure positive coordinates.
- False northing
 - from the equator
 - with a 10,000km false northing for positions south of the equator



UTM Zone



TM2 台灣橫麥卡托二度分帶投影坐標系統

UTM in Taiwan

Coordinate System

Select the Coordinate System to view the available c

Current XY Details

Hu Tzu Shan UTM Zone 51N

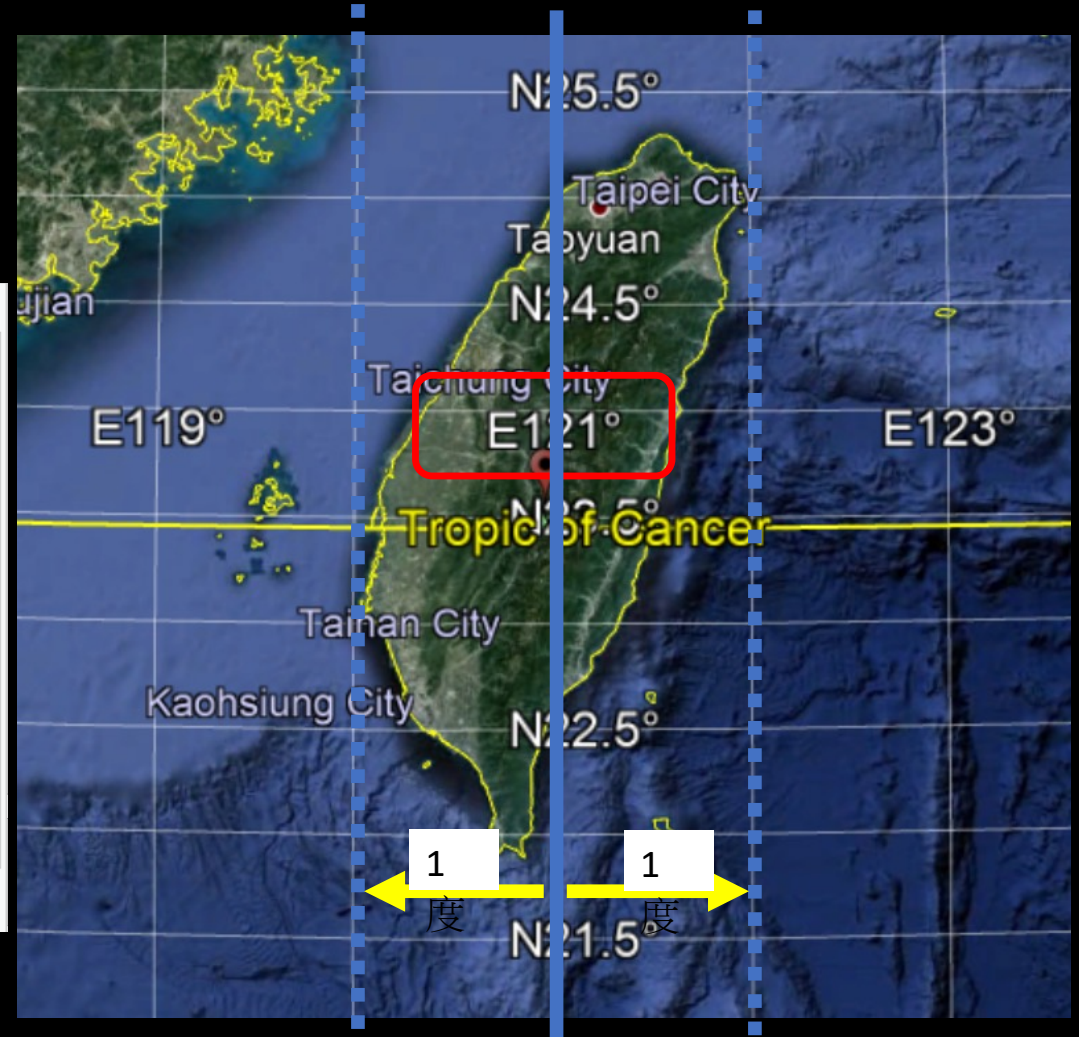
XY Coordinate Systems Available utm

- Fahud UTM Zone 39N
- Fahud UTM Zone 40N
- Hong Kong 1980 UTM Zone 49N
- Hong Kong 1980 UTM Zone 50N
- Hu Tzu Shan UTM Zone 51N
- IGM 1995 UTM Zone 32N
- IGM 1995 UTM Zone 33N
- IGRS UTM Zone 37N

Modify Projected Coordinate System

Name	Hu Tzu Shan UTM Zone 51N_1
Linear Unit	Meters
Meters per unit	1
Projection	Transverse Mercator
False Easting	500000
False Northing	0
Central Meridian	123
Scale Factor	0.9996
Latitude Of Origin	0
Geographic Coordinate System	GCS Hu Tzu Shan
Name	GCS Hu Tzu Shan
Angular Unit	Degree
Radians per unit	0.0174532925199433
Prime Meridian	Greenwich

Save Cancel



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台灣常用的坐標系統及EPSG代碼

 2013年6月30日

 [閱讀評論](#)

 [發表評論](#)

橫麥卡托(Transverse Mercator)系：

TM2 (TWD97, 中央經線121度) (適用臺灣本島, 民國87年之後施行迄今) => **EPSG:3826**

TM2 (TWD97, 中央經線119度) (適用澎湖金門馬祖, 民國87年之後施行迄今) => **EPSG:3825**

TM2 (TWD67, 中央經線121度) (適用臺灣本島, 民國69年之後施行) => **EPSG:3828** (內建參數有誤, 需自行修正)

TM2 (TWD67, 中央經線119度) (適用澎湖金門馬祖, 民國69年之後施行) => **EPSG:3827**

TM3 (TWD67, 中央經線121度) (適用臺灣本島及澎湖, 民國58年之後短暫使用) => 無EPSG代號, 需自行定義

經緯度(Latitude-Longitud)系：

WGS84經緯度 (全球性資料, 如: GPS) => **EPSG:4326**

TWD97經緯度 (國土測繪中心發佈全國性資料) => **EPSG:3824**

TWD67經緯度 (部分地籍圖圖解數化成果) => **EPSG:3821**

虎子山經緯度 (日治時期陸測地形圖) => **EPSG:4236**

其他：

Spherical Mercator (圖磚、WMTS, 如: Google Map) => **EPSG:3857**

虎子山UTM zone 51N (中美合作軍用地形圖) => **EPSG:3829**

<http://gis.rchss.sinica.edu.tw/qgis/?p=2823>

We will have a quiz next week!

